

Chapter 1

Ample and Nef Line Bundles

Revised chapter corresponding to original pp. 6–96.

Original numbering and proof order retained.

The accompanying correction report gives the original page locations.

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In this chapter all schemes will be considered over $\mathbb{C}$ unless another ground field is explicitly specified. Schemes are of finite type over the ground field unless stated otherwise. Some results hold over more general fields; characteristic-zero assumptions are retained where they are needed.

1.1 Cartier Divisors, Line Bundles and Linear Series

1.1A Cartier Divisors and Line Bundles

Cartier divisors are important for studying maps from a given variety to a projective space.

Robin Hartshorne

Definition 1.1.1 (Cartier divisor). A Cartier divisor on X is a global section of the quotient sheaf $\mathcal{K}_X^*/\mathcal{O}_X^*$, where $\mathcal{K}_X$ is the sheaf of total quotient rings:

$$\mathrm{Div}(X) := \Gamma(X, \mathcal{K}_X^*/\mathcal{O}_X^*).$$

Concretely, $D \in \mathrm{Div}(X)$ is represented by a collection $\{(U_i, f_i)\}$, where the U_i cover X , $f_i \in \Gamma(U_i, \mathcal{K}_X^*)$, and on $U_{ij} = U_i \cap U_j$,

$$f_i = g_{ij}f_j \quad \text{for some } g_{ij} \in \Gamma(U_{ij}, \mathcal{O}_X^*).$$

The function f_i is called a local equation for D at any point $x \in U_i$. Two such collections represent the same Cartier divisor if, on a common refinement V_k , their local equations satisfy

$$f_i = h_k f'_k \quad \text{for some } h_k \in \Gamma(V_k, \mathcal{O}_X^*).$$

Although the local equations are multiplied, we use additive notation:

$$D = \{(U_i, f_i)\}, \quad D' = \{(U_i, f'_i)\}, \quad D + D' = \{(U_i, f_i f'_i)\}.$$

Here a common refinement has been chosen. The support of D is the set of points where a local equation is not a unit in $\mathcal{O}_{X,x}$. The divisor is effective if every f_i is regular, hence a nonzerodivisor, on U_i . We write $D \geq 0$, and $D \geq D'$ if $D - D' \geq 0$.

Remark 1.1.2. For a general scheme, $\mathcal{K}_X$ is the sheaf associated to

$$U \mapsto S(U)^{-1}\Gamma(U, \mathcal{O}_X),$$

where $S(U)$ consists of sections whose germs are nonzerodivisors at every point of U . If X is locally Noetherian, then

$$\Gamma(U, \mathcal{K}_X) = A_{\mathrm{tot}} \quad (U = \mathrm{Spec} A \text{ affine}), \quad \mathcal{K}_{X,x} = (\mathcal{O}_{X,x})_{\mathrm{tot}}.$$

In particular, on an integral scheme $\mathcal{K}_X$ is the sheaf of rational functions.

Definition 1.1.3 (Cycles). Let X be a variety, or a scheme, pure of dimension n . A k -cycle on X is a finite $\mathbb{Z}$ -linear combination of integral closed subvarieties of dimension k . The group of these cycles is denoted by $Z_k(X)$. A Weil divisor is an $(n-1)$ -cycle; we write $\text{WDiv}(X) = Z_{n-1}(X)$.

Remark 1.1.4. There is a cycle map

$$\text{Div}(X) \longrightarrow \text{WDiv}(X), \quad D \longmapsto [D] = \sum_V \text{ord}_V(D)[V].$$

Here V runs through the integral codimension-one subvarieties. If X is normal, then $\mathcal{O}_{X, \eta_V}$ is a discrete valuation ring and $\text{ord}_V(D) = v_V(f_i)$ whenever $\eta_V \in U_i$. The compatibility of the local equations makes this independent of i . For a general pure-dimensional Noetherian scheme, the order is defined using lengths: write $f_i = a/b$ in $A = \mathcal{O}_{X, \eta_V}$ with a, b nonzerodivisors, and set

$$\text{ord}_V(D) = \text{length}_A(A/aA) - \text{length}_A(A/bA).$$

One should not call this a discrete valuation without the normality assumption. If X is normal, the cycle map is injective; if X is nonsingular, or more generally locally factorial, it is an isomorphism. Unless otherwise specified, “divisor” will mean Cartier divisor.

Definition 1.1.5. A principal Cartier divisor is $\text{div}(f)$ for $f \in \Gamma(X, \mathcal{K}_X^*)$. Two Cartier divisors are linearly equivalent, written $D_1 \sim_{\text{lin}} D_2$, if

$$D_1 - D_2 = \text{div}(f) \quad \text{for some } f \in \Gamma(X, \mathcal{K}_X^*).$$

Remark 1.1.6 (Pullback of Cartier divisors). Let D be a Cartier divisor on X and let $f : Y \rightarrow X$ be a morphism. One would like to pull back its local equations:

$$D = \{(U_i, f_i)\} \longmapsto f^*D = \{(f^{-1}U_i, f_i \circ f)\}.$$

A sufficient condition, for locally Noetherian schemes, is

$$f(\eta) \notin \text{Supp}(D) \quad \text{for every } \eta \in \text{Ass}(Y).$$

Equivalently, no associated subvariety of Y is mapped into $\text{Supp}(D)$. An associated subvariety may be embedded; it need not be an irreducible component. If Y is reduced, it is enough to check its irreducible components.

For an effective D , this is precisely the condition that the pulled-back local equations be nonzerodivisors. If an associated point maps into D , the corresponding local equation becomes a zerodivisor, so it does not define an effective Cartier divisor on Y . The problem is not merely whether the function vanishes identically on an open set.

Now we discuss the general construction, following [GW20, 11.16].

Inverse Image of Cartier Divisors

We can always pull back line bundles, but we cannot always pull back a specified Cartier divisor. In practice, one often chooses a linearly equivalent representative with suitable support. This is a form of the “moving lemma” philosophy; such a choice still requires justification.

Key Question

Let $f : X \rightarrow Y$ be a morphism of schemes. When does $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ extend to a homomorphism $\mathcal{K}_Y \rightarrow f_*\mathcal{K}_X$? Such an extension induces

$$\mathcal{K}_Y^*/\mathcal{O}_Y^* \longrightarrow f_*(\mathcal{K}_X^*/\mathcal{O}_X^*)$$

and hence a homomorphism

$$f^* : \text{Div}(Y) \longrightarrow \text{Div}(X).$$

Definition 1.1.7 (Pullback). Suppose $\mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ extends to $\mathcal{K}_Y \rightarrow f_*\mathcal{K}_X$. For a Cartier divisor D on Y , its image under the induced map above is called the pullback f^*D .

Proposition 1.1.8. *The pullback has the following properties:*

- (a) $f^*(D_1 + D_2) = f^*D_1 + f^*D_2$;
- (b) $f^*\mathcal{O}_Y(D) \simeq \mathcal{O}_X(f^*D)$.

The pullback of line bundles gives $f^ : \text{Pic}(Y) \rightarrow \text{Pic}(X)$ for every morphism f , even when not all Cartier divisors can be pulled back.*

Proof. We give a sketch. For (a), use $(f_1f_2) \circ f = (f_1 \circ f)(f_2 \circ f)$. For (b), both line bundles are locally generated by the inverse of a pulled-back local equation. $\square$

Theorem 1.1.9 (Pullback Condition). *Let $f : X \rightarrow Y$ be a morphism of schemes. Assume one of the following:*

- (a) *f is flat;*
- (b) *X is reduced and locally Noetherian, and each irreducible component of X dominates an irreducible component of Y .*

Then $f^\flat : \mathcal{O}_Y \rightarrow f_\mathcal{O}_X$ extends to $\mathcal{K}_Y \rightarrow f_*\mathcal{K}_X$.*

Proof. Recall the presheaf in Remark 1.1.2. It is enough to show that a section which is a nonzerodivisor at every point remains so after pullback.

Case (a): flatness preserves the injectivity of multiplication by a nonzerodivisor.

Case (b): on a reduced locally Noetherian scheme, the zerodivisors are the union of the minimal primes. A generic point of an irreducible component of X maps to the generic point of an irreducible component of Y . A regular section on Y is a unit at such a generic point: its local ring has dimension zero, and a nonzerodivisor in a zero-dimensional local ring is a unit. Its pullback is therefore nonzero at every generic point of X , hence is a nonzerodivisor on X . $\square$

Remark 1.1.10 (Divisor of zeros). Let $s \in \Gamma(X, \mathcal{L})$ be a global section of a line bundle. Assume its local equations are nonzerodivisors. For locally Noetherian X , this means that s does not vanish at any associated point. If X is reduced, it is enough that s not vanish identically on any irreducible component. Then $(s)_0 = \text{div}(s)$ is an effective Cartier divisor.

Line Bundles

A Cartier divisor $D = \{(U_i, f_i)\}$ determines the line bundle $\mathcal{O}_X(D) \subseteq \mathcal{K}_X$, with

$$\mathcal{O}_X(D)|_{U_i} = \mathcal{O}_{U_i} f_i^{-1}.$$

Thus, for an open subset $U \subseteq U_i$,

$$\Gamma(U, \mathcal{O}_X(D)) = \{h \in \Gamma(U, \mathcal{K}_X) \mid hf_i \in \Gamma(U, \mathcal{O}_X)\}.$$

The zero section is included. If X is integral, a nonzero rational function h belongs to this space precisely when $\text{div}(h) + D|_U \geq 0$. If $f_i = g_{ij}f_j$, the local frames $e_i = f_i^{-1}$ satisfy $e_j = g_{ij}e_i$ on U_{ij} . This specifies the transition convention.

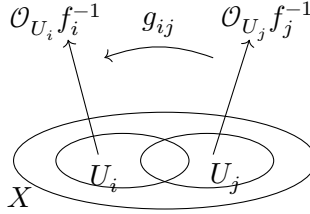


Figure 1.1: Line bundle and local frames

Moreover, the exact sequence of sheaves of abelian groups

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{K}_X^* \longrightarrow \mathcal{K}_X^*/\mathcal{O}_X^* \longrightarrow 1$$

induces the exact sequence

$$\Gamma(X, \mathcal{K}_X^*) \longrightarrow \text{Div}(X) \longrightarrow H^1(X, \mathcal{O}_X^*) \simeq \text{Pic}(X).$$

Consequently, $\mathcal{O}_X(D_1) \simeq \mathcal{O}_X(D_2)$ if and only if $D_1 \sim_{\text{lin}} D_2$. For a Weil divisor D on a normal variety, $\mathcal{O}_X(D)$ denotes the divisorial sheaf; it need not be a line bundle unless D is Cartier.

Example 1.1.11 (Divisor of a section of a line bundle). Let X be an integral variety and $\mathcal{L}$ a line bundle. For a nonzero section $s \in H^0(X, \mathcal{L})$, choose local trivializations $\varphi_i : \mathcal{L}|_{U_i} \rightarrow \mathcal{O}_{U_i}$ and define

$$\operatorname{div}(s) = (s)_0 := \{(U_i, \varphi_i(s|_{U_i}))\}.$$

This is an effective Cartier divisor. If $\mathcal{L} = \mathcal{O}_X(D)$, the local trivialization

$$\mathcal{O}_X(D)|_{U_i} = \mathcal{O}_{U_i} f_i^{-1} \rightarrow \mathcal{O}_{U_i}$$

is multiplication by f_i , so

$$\operatorname{div}(s) = \{(U_i, s|_{U_i} f_i)\}.$$

If D is effective, there is a canonical section $s_D = 1 \in H^0(X, \mathcal{O}_X(D))$, viewed as a rational function. Thus

$$\operatorname{div}(s_D) = \{(U_i, 1 \cdot f_i)\} = D,$$

which recovers the original Cartier divisor.

Now we recall some standard facts in algebraic geometry.

Canonical Bundle and Cartier Divisor

Let X be a smooth projective variety over $\mathbb{C}$ of dimension n . The canonical line bundle is

$$\omega_X := \bigwedge^n \Omega_X^1 = \Omega_X^n.$$

A corresponding Cartier divisor K_X , with $\mathcal{O}_X(K_X) \simeq \omega_X$, is called a canonical divisor. It is determined up to linear equivalence. The symbol K_X here is different from the sheaf $\mathcal{K}_X$ of rational functions.

Kodaira Vanishing Theorem

Let L be an ample Cartier divisor on a smooth projective variety X over a field of characteristic zero. Then

$$H^i(X, \mathcal{O}_X(K_X + L)) = 0 \quad \text{for every } i > 0.$$

We will prove this in Chapter 4, using the covering trick.

1.1B Linear Series

Let X be an integral variety, $\mathcal{L}$ a line bundle on X , and $V \subseteq H^0(X, \mathcal{L})$ a nonzero finite-dimensional subspace. We set

$$|V| := \mathbb{P}_{\text{sub}}(V),$$

the projective space of one-dimensional subspaces of V . When X is complete, its points correspond to the divisors of nonzero sections of V , up to multiplication by a nonzero scalar. This is the linear series of Cartier divisors associated to V . For a complete variety, $H^0(X, \mathcal{L})$ is finite-dimensional [Har77, III, Remark 8.8.1], so $V = H^0(X, \mathcal{L})$ gives the complete linear series. For a Cartier divisor D , we write $|D|$ for the complete linear series of $\mathcal{O}_X(D)$.

Morphism to a Projective Space

Evaluation of sections in V gives an $\mathcal{O}_X$ -linear morphism

$$\text{eval}_V : V \otimes_{\mathbb{C}} \mathcal{O}_X \longrightarrow \mathcal{L}, \quad \sum_i v_i \otimes f_i \mapsto \sum_i f_i v_i|_U$$

on an open set $U \subseteq X$. Here the target is $\mathcal{L}(U)$. At a point x , the value lies in the one-dimensional $\kappa(x)$ -vector space $\mathcal{L}(x) := \mathcal{L}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, identified with $\kappa(x)$ only after choosing a local trivialization.

Definition 1.1.12 (Base ideal). The base ideal of $|V|$ is

$$\mathfrak{b}(|V|) = \mathfrak{b}(X, |V|) \subseteq \mathcal{O}_X,$$

the image of the map $V \otimes_{\mathbb{C}} \mathcal{L}^\vee \rightarrow \mathcal{O}_X$ induced by eval_V and $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^\vee \simeq \mathcal{O}_X$. Here $\mathcal{L}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$, and locally $v \otimes \lambda \mapsto \lambda(v)$.

Definition 1.1.13 (Scheme-theoretic base locus). The base scheme of $|V|$ is the closed subscheme defined by $\mathfrak{b}(|V|)$. Its underlying closed set is

$$\text{Bs}(|V|) := \text{Supp}(\mathcal{O}_X/\mathfrak{b}(|V|)).$$

For a complete linear series, we also write $\mathfrak{b}(|\mathcal{L}|)$ or $\mathfrak{b}(|D|)$.

Remark 1.1.14.

$$\text{Bs}(|V|) = \{x \in X \mid s(x) = 0 \text{ in } \mathcal{L}(x) \text{ for every } s \in V\}.$$

Example 1.1.15. Suppose X is a complete variety and D a Cartier divisor. For integers $m, \ell \geq 1$, there is an inclusion

$$\mathfrak{b}(|\ell D|) \mathfrak{b}(|m D|) \subseteq \mathfrak{b}(|(\ell + m)D|).$$

Indeed, multiplication of sections gives

$$\mu : H^0(X, \mathcal{O}_X(\ell D)) \otimes H^0(X, \mathcal{O}_X(m D)) \longrightarrow H^0(X, \mathcal{O}_X((\ell + m)D)), \quad s \otimes t \mapsto st.$$

Tensoring this map with $\mathcal{O}_X(-(\ell + m)D)$ and then evaluating gives the product of the two base ideals, which is contained in the image of the full evaluation map for $|(\ell + m)D|$.

Definition 1.1.16 (Free linear series). The linear series $|V|$ is free, or base-point-free, if $\text{Bs}(|V|) = \emptyset$, equivalently $\mathfrak{b}(|V|) = \mathcal{O}_X$. A Cartier divisor or line bundle is free if its complete linear series is free. We also say the line bundle is generated by its global sections, or globally generated.

Proposition 1.1.17. *The linear series $|V|$ is free if and only if, for every point $x \in X$, there is a section $s \in V$ such that $s(x) \neq 0$.*

Proof. The condition $\mathfrak{b}(|V|) = \mathcal{O}_X$ is equivalent to the surjectivity of $V \otimes \mathcal{L}_x^\vee \rightarrow \mathcal{O}_{X,x}$ for every x . After trivializing $\mathcal{L}_x$, this says that the local coefficients of the sections generate the unit ideal. In a local ring this holds if and only if at least one coefficient is a unit, equivalently

$$s_x \notin \mathfrak{m}_x \mathcal{L}_x \iff s(x) \neq 0 \text{ in } \mathcal{L}(x)$$

for some $s \in V$. The module $\mathfrak{m}_x \mathcal{L}_x$ is identified with $\mathfrak{m}_x$ only after a trivialization has been chosen. $\square$

Assume now $\dim V \geq 2$, and set $B = \text{Bs}(|V|)$. The linear series determines a morphism

$$\varphi = \varphi_{|V|} : X - B \longrightarrow \mathbb{P}(V).$$

Here, consistently with the quotient convention, $\mathbb{P}(V)$ parametrizes one-dimensional quotients of V ; thus $\mathbb{P}(V) = \mathbb{P}_{\text{sub}}(V^\vee)$. It is different from the projective space $|V| = \mathbb{P}_{\text{sub}}(V)$. For a closed point $x \in X - B$, the corresponding quotient is

$$V \longrightarrow \mathcal{L}(x), \quad s \longmapsto s(x),$$

whose kernel is the hyperplane of sections vanishing at x . At a base point this map is zero and does not define a quotient line.

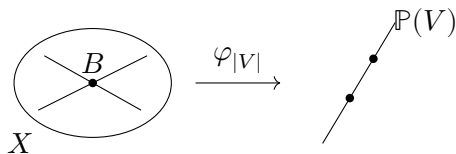


Figure 1.2: The formula for the morphism is undefined on the base locus

For a basis $s_0, \dots, s_r$ of V , the homogeneous-coordinate expression is

$$\varphi : x \longmapsto [s_0(x) : \dots : s_r(x)].$$

Since X is integral and $V \neq 0$, the open set $X - B$ is nonempty and dense. Thus $\varphi_{|V|}$ is a rational map $X \dashrightarrow \mathbb{P}(V)$. If $|V|$ is free, it is a morphism on all of X .

When $B = \emptyset$, the construction can be reversed [Har77, II, Remark 7.8.1]. Suppose $\varphi : X \rightarrow \mathbb{P} = \mathbb{P}(V)$ is a morphism whose image is not contained in a

hyperplane, with $V = H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(1))$. If $\varphi^*s = 0$ for some $s \in V$, the image lies in the hyperplane defined by s , so $s = 0$. Hence

$$\varphi^* : V \hookrightarrow H^0(X, \varphi^*\mathcal{O}_{\mathbb{P}}(1))$$

is injective. Its image generates $\varphi^*\mathcal{O}_{\mathbb{P}}(1)$, and the associated morphism is precisely φ . In particular, if H is a hyperplane which does not contain the image, this line bundle is $\mathcal{O}_X(\varphi^*H)$.

Example 1.1.18 (Projection). Suppose $W \subseteq V$ has dimension at least two. Then

$$\mathfrak{b}(|W|) \subseteq \mathfrak{b}(|V|), \quad \text{Bs}(|V|) \subseteq \text{Bs}(|W|).$$

On $X - \text{Bs}(|W|)$ one has

$$\varphi_{|W|} = \pi \circ \varphi_{|V|}, \quad \pi : \mathbb{P}(V) - \mathbb{P}(V/W) \longrightarrow \mathbb{P}(W),$$

where π is linear projection with centre $\mathbb{P}(V/W)$.

If $|W|$ is free and X is complete, then $|V|$ is free as well. The image $Z = \varphi_{|V|}(X)$ is a closed projective subvariety of $\mathbb{P}(V)$ and avoids the centre. The restricted projection $\pi|_Z : Z \rightarrow \mathbb{P}(W)$ is proper: its source is proper and its target is separated. It is also affine, because the linear projection above is an affine morphism and Z is closed in its domain. Therefore $\pi|_Z$ is finite. Thus the two morphisms differ by a finite projection of $\varphi_{|V|}(X)$. Completeness makes the image closed; it does not assert that $\varphi_{|V|} : X \rightarrow \mathbb{P}(V)$ is a closed immersion.

1.1C Intersection Numbers via the Cohomological Approach

We follow [Deb01, Section 1.2]. In this subsection schemes are of finite type over a field k . When discussing varieties over an algebraically closed field, we use Hartshorne's convention.

For a projective subvariety $X \subseteq \mathbb{P}_k^N$ of dimension n , the function $m \mapsto \chi(X, \mathcal{O}_X(m))$ is a polynomial of degree n . Indeed, for $m \gg 0$, Serre vanishing gives $\chi(X, \mathcal{O}_X(m)) = h^0(X, \mathcal{O}_X(m))$, whose value is the Hilbert polynomial [Har77, I, 7.5; III, 5.2]. Its leading term is

$$P_X(m) = \frac{\deg X}{n!} m^n + O(m^{n-1}).$$

If H is a hyperplane section, $\deg X$ will be denoted by (H^n) . To define $(D_1 \cdots D_n)$ for Cartier divisors on a proper scheme of dimension n , we use Euler characteristics, following [Kle66]; see also [KM98, 1.36], [Deb01, 1.5], [Laz04a, 1.1.24] and [Iit82, Chapter 8].

Theorem 1.1.19 (Asymptotic Riemann–Roch). *Let $D_1, \dots, D_r$ be Cartier divisors on a proper scheme X and $\mathcal{F}$ a coherent sheaf on X . The function*

$$(m_1, \dots, m_r) \mapsto \chi(X, \mathcal{F}(m_1 D_1 + \cdots + m_r D_r))$$

agrees on $\mathbb{Z}^r$ with a polynomial with rational coefficients of total degree at most $\dim \text{Supp } \mathcal{F}$. In particular, for integral X of dimension $n \geq 1$,

$$\begin{aligned}\chi(X, mD) &:= \chi(X, \mathcal{O}_X(mD)) = \frac{(D^n)}{n!} m^n + O(m^{n-1}), \\ \chi(X, \mathcal{F}(mD)) &= \text{rank}(\mathcal{F}) \frac{(D^n)}{n!} m^n + O(m^{n-1}).\end{aligned}$$

Here (D^n) is the self-intersection number defined below, and $\text{rank}(\mathcal{F}) = \dim_{k(X)} \mathcal{F}_\eta$ is the length of the stalk at the generic point. In dimension zero the Euler characteristic is constant.

Proof. We use induction on $\dim \text{Supp } \mathcal{F}$. A coherent sheaf on a Noetherian scheme admits a finite filtration by coherent subsheaves whose successive quotients are torsion-free coherent sheaves on integral closed subschemes. Euler characteristic is additive in exact sequences, so it is enough to consider integral X and torsion-free $\mathcal{F}$.

Choose a nonempty open set U on which $\mathcal{F}$ is free of rank q . Torsion-freeness gives an embedding

$$\mathcal{F} \hookrightarrow \mathcal{K}_X^q,$$

which may be chosen to identify $\mathcal{F}|_U$ with $\mathcal{O}_U^q$. Set $\mathcal{G} = \mathcal{F} \cap \mathcal{O}_X^q$ inside $\mathcal{K}_X^q$. There are exact sequences

$$0 \rightarrow \mathcal{G} \rightarrow \mathcal{F} \rightarrow \mathcal{G}_1 \rightarrow 0, \quad 0 \rightarrow \mathcal{G} \rightarrow \mathcal{O}_X^q \rightarrow \mathcal{G}_2 \rightarrow 0.$$

The coherent sheaves $\mathcal{G}_1$ and $\mathcal{G}_2$ vanish on U , so their supports have smaller dimension. Tensoring both sequences with $\mathcal{O}_X(\sum m_i D_i)$ and using induction reduces the polynomial assertion to $\mathcal{F} = \mathcal{O}_X$. The same calculation also shows that the top-degree part for $\mathcal{F}$ is q times the top-degree part for $\mathcal{O}_X$.

Step 1: First treat a single divisor D , with X integral. Set

$$I_1 = \mathcal{O}_X(-D) \cap \mathcal{O}_X, \quad I_2 = \mathcal{O}_X(D) \cap \mathcal{O}_X$$

inside $\mathcal{K}_X$, and let Y_j be the subscheme defined by I_j . Both Y_j have smaller dimension than X . Since tensoring with a line bundle preserves intersections,

$$I_1(D) = \mathcal{O}_X \cap \mathcal{O}_X(D) = I_2.$$

Thus there are exact sequences

$$\begin{aligned}0 \rightarrow I_1(mD) \rightarrow \mathcal{O}_X(mD) \rightarrow \mathcal{O}_{Y_1}(mD) \rightarrow 0, \\ 0 \rightarrow I_2((m-1)D) \rightarrow \mathcal{O}_X((m-1)D) \rightarrow \mathcal{O}_{Y_2}((m-1)D) \rightarrow 0.\end{aligned}$$

Using $I_1(mD) = I_2((m-1)D)$ gives

$$\chi(X, mD) - \chi(X, (m-1)D) = \chi(Y_1, mD) - \chi(Y_2, (m-1)D).$$

By induction the right side is a rational polynomial of degree at most $\dim X - 1$. A function on $\mathbb{Z}$ whose first difference is such a polynomial agrees, for every integer, with a rational polynomial of degree at most $\dim X$: take a polynomial antidifference, and note that the difference between the two functions is constant on $\mathbb{Z}$. This also handles the zero polynomial. The dimension-zero case is immediate, since tensoring by a line bundle does not change lengths.

Step 2: For several divisors we use the following elementary lemma.

Lemma 1.1.20. *Let $d \geq 0$, and let $f : \mathbb{Z}^r \rightarrow \mathbb{Z}$ be a function which is a polynomial of degree at most d in each variable separately when the others are fixed. Then f agrees on $\mathbb{Z}^r$ with a polynomial in r variables with rational coefficients.*

Proof. Use induction on r . The assertion for $r = 1$ is the hypothesis. For $r > 1$, write

$$f(m_1, \dots, m_r) = \sum_{j=0}^d f_j(m_1, \dots, m_{r-1}) m_r^j,$$

where the coefficient functions take values in $\mathbb{Q}$, not necessarily in $\mathbb{Z}$. Choose distinct integers $c_0, \dots, c_d$. By induction,

$$\sum_{j=0}^d f_j(m_1, \dots, m_{r-1}) c_i^j = P_i(m_1, \dots, m_{r-1}) \quad (0 \leq i \leq d)$$

for rational polynomials P_i . In matrix form,

$$\begin{pmatrix} 1 & c_0 & \cdots & c_0^d \\ 1 & c_1 & \cdots & c_1^d \\ \vdots & \vdots & & \vdots \\ 1 & c_d & \cdots & c_d^d \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_d \end{pmatrix} = \begin{pmatrix} P_0 \\ P_1 \\ \vdots \\ P_d \end{pmatrix}.$$

The Vandermonde matrix is invertible over $\mathbb{Q}$, so every f_j is a rational linear combination of the P_i , hence a polynomial. $\square$

To apply the lemma, fix all but one of the m_i . The one-variable argument applies to the line bundle obtained by the fixed twists, so the degree in that variable is at most $\dim X$. Therefore

$$\chi(X, m_1 D_1 + \cdots + m_r D_r) = P(m_1, \dots, m_r)$$

for a rational polynomial P . If $P \neq 0$ has total degree d , choose integers $n_1, \dots, n_r$ at which its top homogeneous part is nonzero. Then $P(n_1 T, \dots, n_r T)$ has degree d . Setting $D' = n_1 D_1 + \cdots + n_r D_r$, Step 1 shows that this degree is at most $\dim X$. This proves the total-degree bound. The top-coefficient formulas now follow from Definition 1.1.21 and Remark 1.1.22, together with the reduction for a sheaf of rank q above. $\square$

Definition 1.1.21. Let $D_1, \dots, D_r$ be Cartier divisors on a proper scheme X , where $r \geq \dim X$. The intersection number $(D_1 \cdots D_r)$ is the coefficient of $m_1 \cdots m_r$ in the polynomial

$$\chi(X, m_1 D_1 + \cdots + m_r D_r).$$

Proposition 1.1.25 shows that it is an integer. By Theorem 1.1.19, it vanishes when $r > \dim X$.

Remark 1.1.22. If $D_1 = \cdots = D_n = D$ and $\dim X = n$, write $\chi(X, mD) = a_n m^n + O(m^{n-1})$. The coefficient of $m_1 \cdots m_n$ in

$$a_n(m_1 + \cdots + m_n)^n + \text{terms of total degree at most } n-1$$

is $n!a_n$. Hence $(D^n) = n!a_n$, giving the asymptotic Riemann–Roch formula. If X is projective and H is a hyperplane section, Serre vanishing for $\mathcal{O}_X(mH)$ with $m \gg 0$ gives

$$(H^n) = \deg X.$$

This is the degree of the embedded variety X , not the degree of a Cartier divisor viewed as a line bundle on a higher-dimensional variety.

Example 1.1.23. Let X be a smooth projective variety of dimension n , and suppose effective Cartier divisors $D_1, \dots, D_n$ meet properly in finitely many points. Then

$$(D_1 \cdots D_n) = \sum_{x \in D_1 \cap \cdots \cap D_n} \text{length}_{\mathcal{O}_{X,x}} \frac{\mathcal{O}_{X,x}}{(f_1, \dots, f_n)} [\kappa(x) : k].$$

Over an algebraically closed field, this is the number of intersection points counted with multiplicity. For a transverse intersection all multiplicities are one. The same length formula holds whenever the local equations form a regular sequence, for example for a proper intersection on a Cohen–Macaulay pure-dimensional scheme. On a general scheme, merely counting the points or the length of their scheme-theoretic intersection is not sufficient without such a hypothesis.

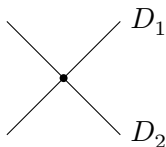


Figure 1.3: Transverse intersection

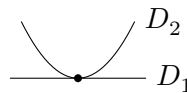


Figure 1.4: Nontransverse intersection

If $Y \subseteq X$ is a proper closed subscheme of dimension at most s , we set

$$(D_1 \cdots D_s \cdot Y) := (\mathcal{O}_X(D_1)|_Y \cdots \mathcal{O}_X(D_s)|_Y).$$

Here the restrictions are restrictions of line bundles; a Cartier divisor itself need not pull back to Y . The polynomial definition applies directly to line bundles. More generally, if $\dim \text{Supp } \mathcal{F} \leq r$, define $(D_1 \cdots D_r \cdot \mathcal{F})$ as the coefficient of $m_1 \cdots m_r$ in $\chi(X, \mathcal{F}(\sum m_i D_i))$.

Example 1.1.24. For a smooth projective integral curve C over an algebraically closed field, Riemann–Roch gives

$$\chi(C, mD) = m \deg(D) + 1 - g(C).$$

Thus $(D \cdot C)$ is the degree of the restricted line bundle. If $C \subseteq X$ is an integral, possibly singular, proper curve and $\tau : \widehat{C} \rightarrow C$ is its normalization, then

$$(D \cdot C) = \deg(\tau^*(\mathcal{O}_X(D)|_C)).$$

This agrees with the preceding polynomial definition: in $0 \rightarrow \mathcal{O}_C \rightarrow \tau_* \mathcal{O}_{\widehat{C}} \rightarrow Q \rightarrow 0$, the cokernel has zero-dimensional support, and twisting does not change its Euler characteristic. Hence the linear coefficients of the two Euler polynomials agree. Over a perfect field $\widehat{C}$ is smooth; over an arbitrary field it is regular, and degree is still defined by Euler characteristic. For a nonreduced or reducible curve, one must also retain the multiplicities of its one-dimensional components.

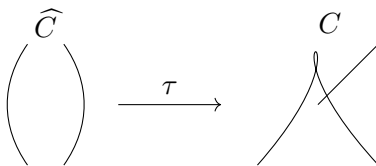


Figure 1.5: Normalization separates the branches

Proposition 1.1.25. Let $D_1, \dots, D_n$ be Cartier divisors on a proper scheme X of dimension n .

- (a) $(D_1, \dots, D_n) \mapsto (D_1 \cdots D_n)$ is multilinear, symmetric and integer-valued.
- (b) If D_n is effective and Y is its associated closed subscheme, then $(D_1 \cdots D_n) = (D_1 \cdots D_{n-1} \cdot Y)$.

Proof. For (a), symmetry follows from the definition. To prove integrality, use the following finite-difference identity. For a polynomial $P(T_1, \dots, T_r)$ of total degree at most r , the coefficient of $T_1 \cdots T_r$ is

$$\sum_{I \subseteq \{1, \dots, r\}} (-1)^{|I|} P(-m_I),$$

where m_I is the indicator vector of I . Indeed, the alternating sum kills every monomial missing a variable, and among monomials of total degree at most r only $T_1 \cdots T_r$ contains all the variables. Consequently,

$$(D_1 \cdots D_r) = \sum_{I \subseteq \{1, \dots, r\}} (-1)^{|I|} \chi \left(X, -\sum_{i \in I} D_i \right) \in \mathbb{Z}.$$

For multilinearity, the same alternating sum with $n+1$ divisors vanishes, since the Euler polynomial has degree at most n . Take the divisors $D_1, D'_1, D_2, \dots, D_n$. Put $S_I = \sum_{i \in I} D_i$ and $\varepsilon_I = (-1)^{|I|}$ for $I \subseteq \{2, \dots, n\}$. Expanding according to the first two divisors gives

$$0 = \sum_I \varepsilon_I (\chi(X, -S_I) - \chi(X, -D_1 - S_I) - \chi(X, -D'_1 - S_I) + \chi(X, -D_1 - D'_1 - S_I)).$$

On the other hand,

$$\begin{aligned} ((D_1 + D'_1)D_2 \cdots D_n) &= \sum_I \varepsilon_I (\chi(X, -S_I) - \chi(X, -D_1 - D'_1 - S_I)), \\ (D_1 D_2 \cdots D_n) &= \sum_I \varepsilon_I (\chi(X, -S_I) - \chi(X, -D_1 - S_I)), \\ (D'_1 D_2 \cdots D_n) &= \sum_I \varepsilon_I (\chi(X, -S_I) - \chi(X, -D'_1 - S_I)). \end{aligned}$$

Subtracting the first expression from the sum of the last two gives zero by the previous identity. This proves additivity in the first variable, and symmetry gives multilinearity.

For (b), fix the index n in the alternating-sum formula. The exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D_n) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

remains exact after twisting by $\mathcal{O}_X(-\sum_{i \in I} D_i)$. Thus

$$\begin{aligned} (D_1 \cdots D_n) &= \sum_{I \subseteq \{1, \dots, n-1\}} (-1)^{|I|} \left[\chi \left(X, -\sum_{i \in I} D_i \right) - \chi \left(X, -D_n - \sum_{i \in I} D_i \right) \right] \\ &= \sum_{I \subseteq \{1, \dots, n-1\}} (-1)^{|I|} \chi \left(Y, -\sum_{i \in I} D_i \right) = (D_1 \cdots D_{n-1} \cdot Y). \end{aligned}$$

□

Projection Formula

Let $\pi : X \rightarrow Y$ be a proper morphism between varieties and let $C \subseteq X$ be an integral proper curve. Define the pushforward cycle by

$$\pi_*[C] = \begin{cases} 0, & \pi(C) \text{ is a point,} \\ d[\pi(C)], & \pi(C) \text{ is a curve,} \end{cases}$$

where $d = [k(C) : k(\pi(C))]$ is the degree, including any inseparable degree.

$$\begin{array}{c} C \\ \left| \right. \\ \xrightarrow{\pi} \end{array} \begin{array}{c} \bullet \\ \pi(C) \end{array}$$

Figure 1.6: A contracted curve

$$\begin{array}{c} C \qquad \pi(C) \\ \left(\begin{array}{c} \xrightarrow{\pi} \\ d = 2 \end{array} \right) \left| \right. \end{array}$$

Figure 1.7: A map of degree two

Corollary 1.1.26 (Projection formula for curves). *If D is a Cartier divisor on Y , then*

$$(\pi^* \mathcal{O}_Y(D) \cdot C) = (D \cdot \pi_*[C]).$$

We also write the left side as $(\pi^ D \cdot C)$, interpreting the pullback as a line bundle when the Cartier divisor itself cannot be pulled back.*

Proof. If $\pi(C)$ is a point, the restricted pullback line bundle is trivial, so both sides are zero. Otherwise, pass to normalizations of the curves. The degree of the induced finite map remains d , and pullback multiplies line-bundle degree by d . Therefore

$$(\pi^* D \cdot C) = d \deg(\mathcal{O}_Y(D)|_{\pi(C)}) = (D \cdot d[\pi(C)]).$$

Here the degree on a singular curve is interpreted as in Example 1.1.24. □

Example 1.1.27 (Expansion of Har77, II, 6.9). A finite dominant morphism $\pi : Y \rightarrow X$ between integral varieties induces a finite extension $\pi^* : k(X) \hookrightarrow k(Y)$. Its degree is $\deg \pi = [k(Y) : k(X)]$. The same definition applies to a dominant generically finite morphism.

The next result comes from [Kle66, p. 299, 6]; see also the proof in Appendix D.6.4 of the notes. For an integral n -dimensional variety Y , the cycle $\pi_*[Y]$ is zero if $\dim \pi(Y) < n$; otherwise it is $[k(Y) : k(\pi(Y))][\pi(Y)]$. Intersection with a cycle is extended linearly.

Proposition 1.1.28. *Let $\pi : Y \rightarrow X$ be a morphism between proper integral varieties and let $D_1, \dots, D_r$ be Cartier divisors on X , where $r \geq \dim Y$. Then*

$$(\pi^* D_1 \cdots \pi^* D_r \cdot Y) = (D_1 \cdots D_r \cdot \pi_*[Y]).$$

The pullbacks on the left are interpreted as line bundles. If π is dominant and generically finite, so that $\dim Y = \dim X$, this becomes

$$(\pi^* D_1 \cdots \pi^* D_r) = \deg(\pi)(D_1 \cdots D_r).$$

In particular, for a birational morphism the intersection numbers agree.

Remark 1.1.29. In Lazarsfeld’s notation, the projection formula for a dominant generically finite morphism between n -dimensional varieties reads

$$\int_Y \pi^* D_1 \cdots \pi^* D_n = \deg(\pi) \int_X D_1 \cdots D_n.$$

For line bundles one writes $\int_X c_1(L_1) \cdots c_1(L_n)$. On a general scheme, the first Chern class is an operator on Chow groups,

$$c_1(L) \cap - : A_k(X) \longrightarrow A_{k-1}(X).$$

It acts on cycle classes modulo rational equivalence; it is not a canonical operation producing a particular representative cycle. Applying n such operators to $[X]$ and taking the degree of the resulting zero-cycle gives the intersection number. See [EH16, 1.30] or [Ful98]. On a smooth projective complex variety, $c_1(L) \in H^2(X, \mathbb{Z})$ also gives the topological description by cup product and integration. Poincaré duality applies to the underlying compact complex manifold; see [GH78].

Numerical Equivalence

Let X be a proper scheme.

Definition 1.1.30. Two Cartier divisors D_1, D_2 are numerically equivalent, written $D_1 \equiv_{\text{num}} D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \quad \text{for every integral curve } C \subseteq X.$$

A divisor is numerically trivial if $D \equiv_{\text{num}} 0$. Such divisors form a subgroup $\text{Num}(X) \subseteq \text{Div}(X)$.

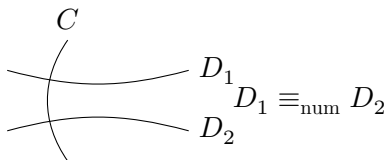


Figure 1.8: Equal intersection numbers with every curve

Definition 1.1.31 (Numerical divisor classes and the Picard number). The group of numerical classes of Cartier divisors is

$$N^1(X) := \text{Div}(X) / \text{Num}(X).$$

It is a free abelian group of finite rank, and its rank $\rho(X) := \text{rank } N^1(X)$ is the Picard number. The usual Néron–Severi group $\text{NS}(X)$ is instead the group of line bundles modulo algebraic equivalence. We keep $N^1(X)$ for numerical classes and do not identify these two groups without qualification. For a smooth projective complex variety, $N^1(X) \simeq \text{NS}(X) / \text{NS}(X)_{\text{tors}}$.

Remark 1.1.32. Equivalently,

$$\rho(X) = \dim_{\mathbb{R}} N^1(X)_{\mathbb{R}}, \quad N^1(X)_{\mathbb{R}} := N^1(X) \otimes_{\mathbb{Z}} \mathbb{R}.$$

We will discuss this real vector space in the section on $\mathbb{R}$ -Cartier divisors.

1.2 The Classical Theory

Question

Given a Cartier divisor D on a projective variety X , what should it mean for D to be positive?

Idea

Ask that $\mathcal{O}_X(D)$ be the hyperplane line bundle for a suitable projective embedding. In this case D is called very ample. If D is effective, it is then a hyperplane section for the embedding defined by the complete linear series. For a specified noneffective representative, it is its linear equivalence class that contains the hyperplane sections.

Good Idea

More conveniently, require some positive multiple of D to be very ample. In this case we say D is ample.

Definition 1.2.1 (Ampleness and very ampleness on a complete scheme). Let X be a complete scheme and L a line bundle on X .

- (i) L is very ample if there is a closed immersion $i : X \hookrightarrow \mathbb{P}^N$ with $L \simeq i^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- (ii) L is ample if $L^{\otimes m}$ is very ample for some integer $m > 0$.

A Cartier divisor D is ample or very ample if $\mathcal{O}_X(D)$ is so.

Example 1.2.2 (Ample line bundles on a curve). For an integral complete curve X and a line bundle L on X ,

$$L \text{ is ample} \iff \deg L > 0.$$

For a nonsingular curve this is [Har77, IV, 3.3]; the criterion also holds for an integral singular curve, using degree as in Example 1.1.24.

Example 1.2.3 (Intersection products). If $D_1, \dots, D_n$ are ample Cartier divisors on an n -dimensional projective variety X , where $n \geq 1$, then $(D_1 \cdots D_n) > 0$. By taking a common positive multiple and using multilinearity, it is enough to treat very ample divisors. Choose general effective representatives successively, avoiding the associated subvarieties of the preceding intersections. They then meet properly, and their intersection through D_{n-1} has a nonzero effective one-cycle

$$[D_1 \cap \cdots \cap D_{n-1}] = \sum_C a_C [C], \quad a_C > 0.$$

Consequently,

$$(D_1 \cdots D_n) = \sum_C a_C \deg(\mathcal{O}_X(D_n)|_C) > 0.$$

We use here the fact, proved below, that an ample line bundle restricts to an ample line bundle on a closed subvariety. For $n = 1$ the assertion is exactly the curve criterion. Arbitrary effective representatives need not meet properly.

1.2A Cohomological Properties

As in [Har77], we already know the following in the projective case. We now state it for a complete scheme, not assumed projective at the outset.

Theorem 1.2.4. *Let L be a line bundle on a complete scheme X . The following are equivalent:*

- (i) L is ample.
- (ii) For every coherent sheaf $\mathcal{F}$ there is an integer $m_1 = m_1(\mathcal{F}, L) > 0$ such that
$$H^i(X, \mathcal{F} \otimes L^{\otimes m}) = 0 \quad (i > 0, m \geq m_1).$$
- (iii) For every coherent sheaf $\mathcal{F}$ there is an integer $m_2 = m_2(\mathcal{F}, L) > 0$ such that $\mathcal{F} \otimes L^{\otimes m}$ is globally generated for every $m \geq m_2$.
- (iv) There is an integer $m_3 = m_3(L) > 0$ such that $L^{\otimes m}$ is very ample for every $m \geq m_3$.

See also [SP25, Tags 0B5T and 01Q3] for the cohomological and generation criteria.

Example 1.2.5 (Sums of Cartier divisors). Let D, E be Cartier divisors on a projective scheme X , with D ample. Choose m_1 so that $m_1 D$ is very ample, and m_2 so that $qD + E$ is free for every $q \geq m_2$. If $m \geq m_1 + m_2$, then

$$mD + E = m_1 D + ((m - m_1)D + E)$$

is the sum of a very ample divisor and a globally generated divisor, hence is very ample [Har77, II, Exercise 7.5(d)]. In particular it is ample. The bound depends on both D and E .

Example 1.2.6 (Ample line bundles on a product). If L and M are ample line bundles on projective schemes X and Y , then $\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M$ is ample on $X \times Y$.

Proof. Choose $N > 0$ such that $L^{\otimes N}$ and $M^{\otimes N}$ are very ample, giving embeddings $i_1 : X \hookrightarrow \mathbb{P}^r$ and $i_2 : Y \hookrightarrow \mathbb{P}^s$. Their product followed by the Segre embedding is

$$i : X \times Y \hookrightarrow \mathbb{P}^r \times \mathbb{P}^s \hookrightarrow \mathbb{P}^{rs+r+s}.$$

By the formula for the Segre hyperplane line bundle,

$$(\mathrm{pr}_1^* L \otimes \mathrm{pr}_2^* M)^{\otimes N} = \mathrm{pr}_1^* L^{\otimes N} \otimes \mathrm{pr}_2^* M^{\otimes N} \simeq i^* \mathcal{O}_{\mathbb{P}^{rs+r+s}}(1).$$

Thus a positive power is very ample, which proves the assertion. $\square$

Proposition 1.2.7 (Finite Pullbacks, I). *Let $f : Y \rightarrow X$ be a finite morphism of complete schemes, and let L be an ample line bundle on X . Then f^*L is ample on Y . In particular, if $Y \subseteq X$ is a closed subscheme, then $L|_Y$ is ample.*

Remark 1.2.8. This is the finite-morphism version of the familiar fact that an ample line bundle restricts to an ample line bundle on a closed subscheme. The assertion can be checked by the cohomological criterion above.

Proof of Proposition 1.2.7. Let $\mathcal{F}$ be coherent on Y . Since f is finite, $f_*\mathcal{F}$ is coherent and higher direct images of quasi-coherent sheaves vanish. The projection formula therefore gives

$$\begin{aligned} H^i(Y, \mathcal{F} \otimes f^*L^{\otimes m}) &\simeq H^i(X, f_*(\mathcal{F} \otimes f^*L^{\otimes m})) \\ &\simeq H^i(X, f_*\mathcal{F} \otimes L^{\otimes m}) = 0 \quad (i > 0, m \gg 0). \end{aligned}$$

Theorem 1.2.4 proves ampleness. A closed immersion is finite, which gives the restriction assertion. $\square$

Corollary 1.2.9 (Globally generated line bundles). *Let X be a complete scheme, let L be globally generated, and let*

$$\varphi = \varphi|_L : X \longrightarrow \mathbb{P}H^0(X, L)$$

be the resulting morphism. The following are equivalent:

- (i) L is ample;
- (ii) φ is finite;
- (iii) $(L \cdot C) = \deg(L|_C) > 0$ for every integral curve $C \subseteq X$.

Proof. If φ is finite, then $L \simeq \varphi^*\mathcal{O}(1)$ is ample by Proposition 1.2.7. An ample L has positive degree on every integral curve by restriction and the curve criterion.

Conversely, φ is proper because X is complete and projective space is separated. If it is not finite, properness and the equivalence “proper and quasi-finite $\iff$ finite” give a positive-dimensional fibre. A positive-dimensional proper variety contains an integral curve C , on which

$$L|_C \simeq (\varphi|_C)^*\mathcal{O}(1) \simeq \mathcal{O}_C.$$

Thus $(L \cdot C) = 0$, contradicting (iii). This proves all the equivalences. The same argument works for any generating linear series defining φ with $L \simeq \varphi^*\mathcal{O}(1)$. $\square$

Proposition 1.2.10. *Let X be a complete scheme and L a line bundle on X .*

- (i) *L is ample on X if and only if L_{red} is ample on X_{red} .*
- (ii) *L is ample on X if and only if it is ample on each irreducible component with its reduced structure.*

Proof. For (i), one direction follows from the finite closed immersion $X_{\text{red}} \hookrightarrow X$. For the converse, let $\mathcal{N}$ be the nilradical, with $\mathcal{N}^r = 0$, and fix a coherent sheaf $\mathcal{F}$. The finite filtration

$$\mathcal{F} \supseteq \mathcal{N}\mathcal{F} \supseteq \dots \supseteq \mathcal{N}^r\mathcal{F} = 0$$

has quotients

$$Q_i = \mathcal{N}^i\mathcal{F} / \mathcal{N}^{i+1}\mathcal{F} = \mathcal{N}^i\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\text{red}}},$$

which are coherent on X_{red} . Hence

$$H^j(X, Q_i \otimes L^{\otimes m}) = H^j(X_{\text{red}}, Q_i \otimes L_{\text{red}}^{\otimes m}) = 0 \quad (j > 0, m \gg 0).$$

Choose a single bound for the finitely many Q_i . Twist $0 \rightarrow \mathcal{N}^{i+1}\mathcal{F} \rightarrow \mathcal{N}^i\mathcal{F} \rightarrow Q_i \rightarrow 0$ by $L^{\otimes m}$. Decreasing induction on i then gives vanishing for $\mathcal{F} \otimes L^{\otimes m}$, and Theorem 1.2.4 applies.

For (ii), by (i) assume X is reduced, and write $X = X_1 \cup \dots \cup X_r$. Restriction gives the forward implication. For the converse, let I be the ideal of X_1 and J the ideal of $X' = X_2 \cup \dots \cup X_r$, both with their reduced structures. Since X is reduced, $I \cap J = 0$, hence $IJ = 0$. In the sequence

$$0 \longrightarrow I\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/I\mathcal{F} \longrightarrow 0,$$

the first sheaf is annihilated by J and is therefore a coherent sheaf on X' , whereas the last is a coherent sheaf on X_1 . Induction on the number of components gives the desired cohomology vanishing for both end terms after twisting by $L^{\otimes m}$. It then gives vanishing for the middle term, proving ampleness. The first sheaf need not vanish at points of $X_1 \cap X'$; its support is contained in X' . $\square$

The next result can be found in [GD61a, 4.7.1] or [KM98, 1.41].

Theorem 1.2.11 (Ampleness in families). *Let $f : X \rightarrow T$ be a proper morphism of Noetherian schemes and let L be a line bundle on X . For $t \in T$, write*

$$X_t = X \times_T \text{Spec } \kappa(t), \quad L_t = L|_{X_t}.$$

If L_0 is ample on X_0 for a point $0 \in T$, then there is an open neighbourhood U of 0 such that L_t is ample on X_t for every $t \in U$. No flatness assumption on f is needed.

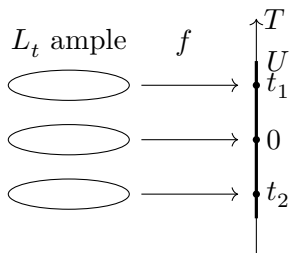


Figure 1.9: Ampleness holds on fibres over an open neighbourhood

Proof. We keep the three steps of the argument. First work over the local base $\text{Spec } A = \text{Spec } \mathcal{O}_{T,0}$, where $\mathfrak{m}$ is the maximal ideal and $A/\mathfrak{m} = \kappa(0)$. In Step 3 we extend the finitely many sections used to an actual open neighbourhood in the original T . This permits 0 to be any point, not necessarily a closed point of T .

Step 1: We claim that for every coherent sheaf $\mathcal{F}$ there is an integer $m(\mathcal{F})$ such that, over this local base,

$$R^i f_*(\mathcal{F} \otimes L^{\otimes m}) = 0 \quad (i \geq 1, m \geq m(\mathcal{F})). \quad (1.1)$$

Use decreasing induction on the cohomological index, with the assertion at each index made for all coherent sheaves. It starts above a bound for the cohomological dimension of f . Choose generators $u_1, \dots, u_p$ of $\mathfrak{m}$. Pulling back $A^p \xrightarrow{u} A \rightarrow \kappa(0) \rightarrow 0$ and tensoring with $\mathcal{F}$ gives

$$\mathcal{F}^p \xrightarrow{u} \mathcal{F} \rightarrow \mathcal{F}_0 \rightarrow 0, \quad \mathcal{F}_0 = \mathcal{F} \otimes \mathcal{O}_{X_0}.$$

Let K be the kernel of the first map and I its image. Thus

$$0 \rightarrow K \rightarrow \mathcal{F}^p \rightarrow I \rightarrow 0, \quad 0 \rightarrow I \rightarrow \mathcal{F} \rightarrow \mathcal{F}_0 \rightarrow 0.$$

Suppose the assertion is known at index $i \geq 2$. For $m \gg 0$, the induction hypothesis gives $R^i f_*(K \otimes L^{\otimes m}) = 0$, and ampleness of L_0 gives

$$H^{i-1}(X_0, \mathcal{F}_0 \otimes L_0^{\otimes m}) = 0.$$

The two long exact sequences yield a surjection

$$M_m^{\oplus p} \xrightarrow{(u_1, \dots, u_p)} M_m, \quad M_m = H^{i-1}(X, \mathcal{F} \otimes L^{\otimes m}).$$

This is a finite A -module by properness. The image is $\mathfrak{m}M_m$, so $M_m = \mathfrak{m}M_m$, and Nakayama's lemma gives $M_m = 0$. Since T is affine, this is the required higher-direct-image vanishing at index $i - 1$. We only use the displayed map given by the u_j ; no assertion that cohomology commutes with tensoring by $A/\mathfrak{m}$ is needed.

Step 2: We next show that, for $m \gg 0$, the evaluation morphism

$$\rho_m : f^* f_* L^{\otimes m} \longrightarrow L^{\otimes m}$$

is surjective. Apply Step 1 to the ideal sheaf $\mathcal{I}_{X_0/X}$. The sequence $0 \rightarrow \mathcal{I}_{X_0/X} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_{X_0} \rightarrow 0$ then gives a surjection

$$H^0(X, L^{\otimes m}) \longrightarrow H^0(X_0, L_0^{\otimes m}) \quad (m \gg 0). \quad (1.2)$$

All sufficiently large powers of L_0 are globally generated. Composing this restriction with evaluation on X_0 , over the field $\kappa(0)$, shows that ρ_m is surjective along X_0 , by Nakayama's lemma. Its cokernel is coherent. The image under the proper map f of the support of this cokernel is closed and misses the closed point of $\text{Spec } A$. Every nonempty closed subset of the spectrum of a local ring contains its closed point, so this support is empty. Thus ρ_m is surjective on the local-base family.

Step 3: Fix one sufficiently large integer m . On the affine local base, the coherent module $f_* L^{\otimes m}$ is generated by finitely many sections $s_0, \dots, s_r$. Composing their pullback with ρ_m gives a surjection $\mathcal{O}_X^{r+1} \rightarrow L^{\otimes m}$.

Return to an affine neighbourhood of 0 in the original T . Sections of a quasi-coherent sheaf on the quasi-compact separated scheme X commute with localization on this base. The finitely many s_i therefore extend after shrinking that neighbourhood once. The failure of these extended sections to generate has closed support, whose proper image misses 0; remove this image. The same sections now give a morphism over the resulting neighbourhood U ,

$$\varphi : X_U \longrightarrow \mathbb{P}_U^r, \quad L^{\otimes m}|_{X_U} \simeq \varphi^* \mathcal{O}_{\mathbb{P}_U^r}(1).$$

The integer m remains fixed throughout these shrinkings.

On X_0 , the morphism φ_0 is finite. Indeed, a positive-dimensional fibre would contain a curve C , and the pullback formula would give $m \deg(L_0|_C) = 0$, contrary to ampleness of L_0 . The locus in X_U where φ is not quasi-finite is closed and disjoint from X_0 . Its image under the proper map f is closed; remove that image from U . Then φ is quasi-finite and proper, hence finite. For each $t \in U$, Proposition 1.2.7 now shows that $L_t^{\otimes m} \simeq \varphi_t^* \mathcal{O}_{\mathbb{P}_{\kappa(t)}^r}(1)$ is ample, and therefore L_t is ample. $\square$

Example 1.2.12 (Asymptotic Riemann–Roch in the ample case). Let D be ample on an integral projective variety X of dimension $n \geq 1$. Then

$$h^0(X, \mathcal{O}_X(mD)) = \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

More generally, for a coherent sheaf $\mathcal{F}$,

$$h^0(X, \mathcal{F} \otimes \mathcal{O}_X(mD)) = \text{rank}(\mathcal{F}) \frac{(D^n)}{n!} m^n + O(m^{n-1}).$$

This follows immediately from Theorem 1.1.19 and Serre vanishing. The zero-dimensional case is constant.

Example 1.2.13 (Upper bounds on h^0). If E is any Cartier divisor on an integral projective variety X of dimension n , then there is a constant $C > 0$ such that

$$h^0(X, \mathcal{O}_X(mE)) \leq Cm^n \quad (m \gg 0).$$

Proof. Fix an ample divisor D . For a fixed sufficiently large a , $\mathcal{O}_X(aD - E)$ has a nonzero global section s , so $aD - E$ is linearly equivalent to an effective divisor. Since X is integral, multiplication by s^m gives an injection

$$\mathcal{O}_X(mE) \hookrightarrow \mathcal{O}_X(maD), \quad t \mapsto s^m t.$$

Taking global sections and using the preceding example gives

$$h^0(X, mE) \leq h^0(X, maD) = \frac{(aD)^n}{n!} m^n + O(m^{n-1}) \leq Cm^n.$$

For $n = 0$ the bound is immediate from finite length. $\square$

Example 1.2.14 (Resolutions of a sheaf). Let D be ample on a projective variety X . Any coherent sheaf $\mathcal{F}$ admits a resolution

$$\cdots \rightarrow \mathcal{O}_X(-p_1 D)^{\oplus a_1} \rightarrow \mathcal{O}_X(-p_0 D)^{\oplus a_0} \rightarrow \mathcal{F} \rightarrow 0$$

for suitable integers $0 \ll p_0 \ll p_1 \ll \cdots$ and finite ranks a_i . The resolution need not terminate.

Proof. Choose $p_0 \gg 0$ such that $\mathcal{F}(p_0 D)$ is globally generated, and choose finitely many generating sections. After twisting back, this gives a surjection $\mathcal{O}_X(-p_0 D)^{\oplus a_0} \rightarrow \mathcal{F}$. Its kernel is coherent, so the same construction applies to the kernel with a sufficiently large p_1 . Repeat this process. $\square$

1.2B Numerical Properties

The ampleness of Cartier divisors can be characterized numerically.

Theorem 1.2.15 (Nakai–Moishezon–Kleiman criterion). *Let L be a line bundle on a projective scheme X . Then L is ample if and only if*

$$\int_V c_1(L)^{\dim V} = (L^{\dim V} \cdot V) > 0$$

for every positive-dimensional integral closed subvariety $V \subseteq X$. In particular, a Cartier divisor D is ample if and only if $(D^{\dim V} \cdot V) > 0$ for all such V .

Before giving a proof, we derive two important consequences. First, ampleness depends only on the numerical equivalence class.

Corollary 1.2.16 (Numerical nature of ampleness). *If D_1, D_2 are numerically equivalent Cartier divisors on a projective scheme X , then D_1 is ample if and only if D_2 is ample.*

We need a lemma to prove this corollary.

Lemma 1.2.17. *Let X be a complete scheme, and suppose $D_i \equiv_{\text{num}} D'_i$ for Cartier divisors $D_1, \dots, D_k, D'_1, \dots, D'_k$. Then*

$$(D_1 \cdots D_k \cdot V) = (D'_1 \cdots D'_k \cdot V)$$

for every integral closed subvariety $V \subseteq X$ of dimension k .

Proof. For $E \equiv_{\text{num}} 0$, the class $c_1(\mathcal{O}_X(D_2)) \cdots c_1(\mathcal{O}_X(D_k)) \cap [V]$ in $A_1(X)$ is represented by a one-cycle γ . Thus

$$(ED_2 \cdots D_k \cdot V) = (E \cdot \gamma) = 0.$$

Apply this to $E = D_1 - D'_1$, then replace the other divisors one at a time. Multilinearity proves the lemma. We use Chow groups and Chern class operators here; X need not have a Chow ring as a smooth variety does. $\square$

Proof of Corollary 1.2.16. The lemma gives $(D_1^{\dim V} \cdot V) = (D_2^{\dim V} \cdot V)$ for every V . Theorem 1.2.15 therefore applies to the two divisors simultaneously. $\square$

Definition 1.2.18 (Ample classes). A numerical class $\delta \in N^1(X)$ is ample if it is the class of an ample Cartier divisor.

Example 1.2.19 (Varieties with Picard number one). If X is a projective variety with $\rho(X) = 1$, then any nonzero effective Cartier divisor D is ample. Indeed, fix an ample Cartier divisor H . The positivity $(D \cdot H^{n-1}) > 0$ shows that $[D]$ and $[H]$ lie on the same half-line in $N^1(X)_{\mathbb{R}}$. Since these are integral classes in a rank-one group, there are positive integers a, b with $aD \equiv_{\text{num}} bH$. By Corollary 1.2.16, aD is ample, and so is D .

We next discuss the converse to finite pullback. In the implication below, the assumption that f^*L is ample already forces f to be finite. Thus finiteness need not be imposed separately.

Corollary 1.2.20 (Finite Pullbacks, II). *Let $f : Y \rightarrow X$ be a surjective morphism of projective schemes, and let L be a line bundle on X . Then*

$$f^*L \text{ ample on } Y \implies L \text{ ample on } X.$$

Proof. A positive-dimensional fibre would contain an integral curve C , on which f^*L is trivial. This contradicts ampleness. Hence f is quasi-finite, and since it is proper it is finite.

Let $V \subseteq X$ be a positive-dimensional integral subvariety. There is an irreducible component W of $f^{-1}(V)$ which dominates V , and we give W its reduced structure. The restricted map $f|_W : W \rightarrow V$ is finite and surjective, so $\dim W = \dim V$ and

$$0 < \int_W c_1(f^*L)^{\dim V} = \deg(W \rightarrow V) \int_V c_1(L)^{\dim V}.$$

Theorem 1.2.15 proves the result. No hyperplane cuts are needed here. For a general projective surjective map, hyperplane cuts can produce a subvariety generically finite over V ; they do not in general guarantee finiteness over every point of V . $\square$

Now we return to the proof of the numerical criterion.

Proof of Theorem 1.2.15. The forward implication follows by choosing $m > 0$ such that $L^{\otimes m}$ is very ample. For the resulting embedding and every V ,

$$m^{\dim V} \int_V c_1(L)^{\dim V} = \deg(V \text{ in this embedding}) > 0.$$

For the converse, use Proposition 1.2.10 to reduce to an integral projective variety X , and write $L \simeq \mathcal{O}_X(D)$. We induct on $n = \dim X$. In dimension zero every line bundle is ample. In dimension one the assertion is the curve criterion. Assume the result in dimensions less than n .

Step 1: We claim that $H^0(X, \mathcal{O}_X(mD)) \neq 0$ for $m \gg 0$. By asymptotic Riemann–Roch,

$$\chi(X, mD) = \frac{(D^n)}{n!} m^n + O(m^{n-1}), \quad (D^n) > 0.$$

Fix a very ample H , and then fix an integer $a \gg 0$ such that $aH + D$ and aH are both very ample. Choose effective representatives $A \sim_{\text{lin}} aH + D$ and $B \sim_{\text{lin}} aH$, so $D \sim_{\text{lin}} A - B$. These divisors are fixed while m tends to infinity. Their defining sections give exact sequences

$$\begin{aligned} 0 \rightarrow \mathcal{O}_X(mD - B) &\xrightarrow{\cdot A} \mathcal{O}_X((m+1)D) \rightarrow \mathcal{O}_A((m+1)D) \rightarrow 0, \\ 0 \rightarrow \mathcal{O}_X(mD - B) &\xrightarrow{\cdot B} \mathcal{O}_X(mD) \rightarrow \mathcal{O}_B(mD) \rightarrow 0. \end{aligned}$$

All integral subvarieties of A and B satisfy the required positivity. They have smaller dimension, so induction makes $\mathcal{O}_A(D)$ and $\mathcal{O}_B(D)$ ample. The higher cohomology of the right-hand terms vanishes for $m \gg 0$. Thus, for $i \geq 2$,

$$H^i(X, mD) \simeq H^i(X, mD - B) \simeq H^i(X, (m+1)D).$$

The dimensions $h^i(X, mD)$ are eventually constant for $i \geq 2$. Hence

$$\chi(X, mD) = h^0(X, mD) - h^1(X, mD) + c$$

for some constant c , possibly zero. Since $\chi(X, mD) \rightarrow +\infty$, we have $h^0(X, mD) \rightarrow +\infty$. This proves the claim. Choose a nonzero section of one such positive multiple and replace D by its effective zero divisor. This is a replacement by a linearly equivalent representative of a positive multiple, and does not change the ampleness question.

Step 2: We show that $\mathcal{O}_X(mD)$ is globally generated for $m \gg 0$. Outside $\text{Supp } D$, the canonical section s_D^m does not vanish. It remains to lift enough sections from the subscheme D . Use

$$0 \rightarrow \mathcal{O}_X((m-1)D) \rightarrow \mathcal{O}_X(mD) \rightarrow \mathcal{O}_D(mD) \rightarrow 0.$$

Induction makes $\mathcal{O}_D(D)$ ample. Thus for $m \gg 0$, $\mathcal{O}_D(mD)$ is globally generated and $H^1(D, \mathcal{O}_D(mD)) = 0$. The long exact sequence gives a surjection

$$H^1(X, (m-1)D) \rightarrow H^1(X, mD).$$

The nonnegative integers $h^1(X, mD)$ are eventually nonincreasing, hence eventually constant. The displayed surjections are then isomorphisms, so exactness implies that

$$H^0(X, mD) \rightarrow H^0(D, \mathcal{O}_D(mD))$$

is surjective. Therefore there are no base points on D , and the canonical section takes care of the complement. This proves global generation. We did not assume $H^1(X, mD) = 0$.

Step 3: For every integral curve $C \subseteq X$, the hypothesis gives $(mD \cdot C) > 0$. Apply Corollary 1.2.9 to the globally generated line bundle $\mathcal{O}_X(mD)$: it is ample, and hence $\mathcal{O}_X(D)$ is ample. $\square$

Remark 1.2.21 (Nakai's criterion on proper schemes). The statement remains true for a complete scheme not assumed projective. Projectivity was used above to write D as the difference of two very ample divisors. The following argument replaces that step by the two ideal sheaves already used in the proof of Theorem 1.1.19. See [Deb01, Section 1.2] and [KM98, 1.37].

Theorem 1.2.22 (Nakai–Moishezon criterion). *A Cartier divisor D on a proper scheme X is ample if and only if*

$$(D^{\dim Y} \cdot Y) > 0$$

for every positive-dimensional integral closed subscheme $Y \subseteq X$. One may include zero-dimensional integral subschemes as well, since their degrees over the ground field are automatically positive.

Proof. If D is ample, a positive multiple defines a projective embedding and

$$m^{\dim Y}(D^{\dim Y} \cdot Y) = \deg(Y \text{ in this embedding}) > 0.$$

Conversely, by Proposition 1.2.10 it is enough to treat integral X . Induct on $n = \dim X$; the zero-dimensional case is immediate. Choose the usual inclusion $\mathcal{O}_X(D) \subseteq \mathcal{K}_X$ and set

$$I_1 = \mathcal{O}_X(-D) \cap \mathcal{O}_X, \quad I_2 = \mathcal{O}_X(D) \cap \mathcal{O}_X.$$

The subschemes Y_j defined by I_j have smaller dimension, and $I_1(D) = I_2$. By induction, $\mathcal{O}_X(D)|_{Y_j}$ is ample. In the exact sequences

$$\begin{aligned} 0 \rightarrow I_1(mD) \rightarrow \mathcal{O}_X(mD) \rightarrow \mathcal{O}_{Y_1}(mD) \rightarrow 0, \\ 0 \rightarrow I_2((m-1)D) \rightarrow \mathcal{O}_X((m-1)D) \rightarrow \mathcal{O}_{Y_2}((m-1)D) \rightarrow 0, \end{aligned}$$

the right-hand terms have vanishing higher cohomology for $m \gg 0$. Consequently, for $i \geq 2$,

$$h^i(X, mD) = h^i(X, I_1(mD)) = h^i(X, I_2((m-1)D)) = h^i(X, (m-1)D).$$

These dimensions are eventually constant. Since $(D^n) > 0$ and $n > 0$, asymptotic Riemann–Roch shows that $\chi(X, mD) \rightarrow +\infty$. It follows that $h^0(X, mD) - h^1(X, mD)$, and therefore $h^0(X, mD)$, tends to infinity. A nonzero section makes some positive multiple of D linearly equivalent to an effective divisor. Replace D by this divisor.

By induction $\mathcal{O}_D(D)$ is ample. The sequence

$$0 \rightarrow \mathcal{O}_X((m-1)D) \rightarrow \mathcal{O}_X(mD) \rightarrow \mathcal{O}_D(mD) \rightarrow 0$$

gives, for $m \gg 0$, surjections $H^1(X, (m-1)D) \rightarrow H^1(X, mD)$. Their dimensions eventually stabilize, so $H^0(X, mD) \rightarrow H^0(D, \mathcal{O}_D(mD))$ is surjective. The right-hand line bundle is globally generated, while s_D^m is nonvanishing outside $\text{Supp } D$. Thus $\mathcal{O}_X(mD)$ is globally generated on all of X .

It defines a proper morphism $\varphi : X \rightarrow \mathbb{P}^r$. A positive-dimensional fibre would contain a curve C and give $\mathcal{O}_X(mD)|_C \simeq \mathcal{O}_C$, contrary to $(D \cdot C) > 0$. Thus φ is finite, and $\mathcal{O}_X(mD) \simeq \varphi^*\mathcal{O}(1)$ is ample by Proposition 1.2.7. Therefore D is ample. $\square$

Example 1.2.23 (Further characterization of ampleness). Let X be a projective scheme and D a Cartier divisor on X . Then D is ample if and only if for every positive-dimensional integral subvariety $V \subseteq X$, there are an integer $m = m(V) > 0$ and a nonzero section $s_V \in H^0(V, \mathcal{O}_V(mD))$ which vanishes at some point of V . This characterization is due to Kleiman [Kle66].

Proof. If D is ample, a sufficiently large multiple is very ample on V , and a hyperplane through a point of V but not containing V supplies the required section.

For the converse, Proposition 1.2.10 reduces to integral X . In dimension zero every line bundle is ample. On an integral curve, the zero divisor of s_V is nonempty and effective, so $m \deg(D|_V) > 0$; the curve criterion applies. Now induct on dimension. Use $V = X$ and replace D by the nonzero effective divisor of s_X , which is linearly equivalent to a positive multiple of the original divisor. The condition is preserved by this replacement, since powers of a section remain nonzero and still vanish at the same point.

Every integral subvariety of the subscheme D satisfies the hypothesis, so induction and Proposition 1.2.10 make $\mathcal{O}_D(D)$ ample. The same exact-sequence and stabilization argument as in Theorem 1.2.15, Step 2, makes $\mathcal{O}_X(mD)$ globally generated for $m \gg 0$. The curve case already gives $(D \cdot C) > 0$ for every integral curve $C \subseteq X$. Corollary 1.2.9 proves ampleness. $\square$

Example 1.2.24 (Growth of cohomology). Let X be a projective scheme of dimension n and D a Cartier divisor. For every coherent sheaf $\mathcal{F}$ and every $i \geq 0$,

$$h^i(X, \mathcal{F}(mD)) = O(m^n) \quad (m \rightarrow +\infty).$$

Proof. Write $D \sim_{\text{lin}} A - B$ with A, B very ample effective divisors chosen so that neither contains an associated subvariety of $\mathcal{F}$. Such choices exist over the infinite ground field. The corresponding multiplication maps are injective, giving

$$\begin{aligned} 0 \rightarrow \mathcal{F}(mD - B) &\xrightarrow{\cdot A} \mathcal{F}((m+1)D) \rightarrow \mathcal{F}|_A((m+1)D) \rightarrow 0, \\ 0 \rightarrow \mathcal{F}(mD - B) &\xrightarrow{\cdot B} \mathcal{F}(mD) \rightarrow \mathcal{F}|_B(mD) \rightarrow 0. \end{aligned}$$

Induct on dimension. For $n = 0$, higher cohomology vanishes and h^0 is bounded by the fixed length of $\mathcal{F}$. For $n > 0$, both restricted sheaves have support of dimension at most $n - 1$. Put $b_m = h^i(X, \mathcal{F}(mD - B))$. The two long exact sequences give

$$\begin{aligned} -h^{i-1}(A, \mathcal{F}|_A((m+1)D)) &\leq h^i(X, \mathcal{F}((m+1)D)) - b_m \\ &\leq h^i(A, \mathcal{F}|_A((m+1)D)), \\ -h^{i-1}(B, \mathcal{F}|_B(mD)) &\leq h^i(X, \mathcal{F}(mD)) - b_m \\ &\leq h^i(B, \mathcal{F}|_B(mD)), \end{aligned}$$

where $h^{-1} = 0$ when $i = 0$. Induction bounds the outer terms by $O(m^{n-1})$. Subtracting the two middle expressions therefore gives

$$|h^i(X, \mathcal{F}((m+1)D)) - h^i(X, \mathcal{F}(mD))| = O(m^{n-1}).$$

Summing this estimate from a fixed starting integer to m gives $h^i(X, \mathcal{F}(mD)) = O(m^n)$. $\square$

1.3 $\mathbb{Q}$ -Cartier Divisors and $\mathbb{R}$ -Cartier Divisors

Goal

Make a small perturbation of a Cartier divisor class and see what happens.

1.3A Definition of $\mathbb{Q}$ -Cartier Divisors

A $\mathbb{Q}$ -Cartier divisor is a rational linear combination of integral Cartier divisors.

Definition 1.3.1 ($\mathbb{Q}$ -Cartier divisors). For a variety or scheme X , set

$$\mathrm{Div}_{\mathbb{Q}}(X) := \mathrm{Div}(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

A $\mathbb{Q}$ -Cartier divisor is an element of this vector space, represented by a finite sum $D = \sum_i c_i A_i$, with $c_i \in \mathbb{Q}$ and $A_i \in \mathrm{Div}(X)$. Clearing denominators gives $D = A/r$ for an integer $r > 0$ and an integral Cartier divisor A . The map $\mathrm{Div}(X) \rightarrow \mathrm{Div}_{\mathbb{Q}}(X)$ kills torsion: for $c \in \mathbb{Q} \setminus \{0\}$, the element cA is zero if and only if A is torsion in $\mathrm{Div}(X)$. The divisor D is effective if it admits an expression $D = \sum_i c_i A_i$ with $c_i \geq 0$ and all A_i effective integral Cartier divisors.

Definition 1.3.2 (Supports). A closed subset $E \subseteq X$ supports $D \in \mathrm{Div}_{\mathbb{Q}}(X)$ if there is an expression $D = \sum_i c_i A_i$ such that

$$\bigcup_i \mathrm{Supp} A_i \subseteq E.$$

Such an expression, and such a supporting set, need not be unique. On a normal variety, the associated Weil $\mathbb{Q}$ -divisor has a canonical support, obtained from its nonzero coefficients along prime divisors. On a general scheme we will specify a supporting set when it is needed.

The usual operations extend to $\mathbb{Q}$ -Cartier divisors.

Definition 1.3.3 (Equivalences and operations on $\mathbb{Q}$ -Cartier divisors). Assume X is complete.

- (i) For a closed subscheme $V \subseteq X$ of dimension k , the intersection product

$$\mathrm{Div}_{\mathbb{Q}}(X)^k \longrightarrow \mathbb{Q}, \quad (D_1, \dots, D_k) \longmapsto (D_1 \cdots D_k \cdot V)$$

is defined by extension of scalars from the integral intersection product.

- (ii) Divisors D_1, D_2 are numerically equivalent, written $D_1 \equiv_{\mathrm{num}} D_2$ or $D_1 \equiv_{\mathrm{num}, \mathbb{Q}} D_2$, if their intersection numbers with every integral curve agree. The vector space of numerical classes is denoted by $N^1(X)_{\mathbb{Q}}$.

- (iii) Write $D_1 \sim_{\mathbb{Q}} D_2$ if a positive integral multiple of $D_1 - D_2$ is the image of a principal Cartier divisor in $\text{Div}_{\mathbb{Q}}(X)$. Equivalently, after a common positive multiple, integral representatives can be chosen which are linearly equivalent in the usual sense. This is $\mathbb{Q}$ -linear equivalence; we reserve $\sim_{\text{lin}}$ for ordinary linear equivalence of integral Cartier divisors when a distinction matters.
- (iv) If $f : Y \rightarrow X$ is a morphism and no associated subvariety of Y maps into a chosen supporting set for D , the local equations can be pulled back and extended $\mathbb{Q}$ -linearly to define f^*D . This is the condition of Remark 1.1.6, applied to the chosen representatives.
- (v) For an arbitrary morphism of complete varieties, or of projective schemes, pullback of line bundles induces a functorial map

$$f^* : N^1(X)_{\mathbb{Q}} \longrightarrow N^1(Y)_{\mathbb{Q}}.$$

It agrees with divisor pullback whenever the latter is defined, because the projection formula shows that numerical equivalence is preserved. Here line bundles on the source have Cartier representatives. For arbitrary proper schemes without this property, the same functorial statement is formulated using numerical classes of line bundles; one should not infer it solely by extending an everywhere-defined Cartier-divisor pullback, which may not exist.

Remark 1.3.4. The operations are computed from $D = \sum_i c_i A_i$ by linearity. In particular,

$$D \equiv_{\text{num}} 0 \iff \sum_i c_i (A_i \cdot C) = 0 \text{ for every curve } C.$$

There is an isomorphism

$$N^1(X)_{\mathbb{Q}} \simeq N^1(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Indeed, write a numerically trivial rational divisor as A/r with $r > 0$. Then A is numerically trivial. Thus its kernel is exactly the rational span of numerically trivial integral Cartier divisors.

Remark 1.3.5. Two integral Cartier divisors in distinct ordinary linear equivalence classes can become $\mathbb{Q}$ -linearly equivalent. Their difference then has torsion class in $\text{Pic}(X)$. Numerical equivalence of integral divisors does not change after extension to $\mathbb{Q}$.

Continue to assume X is complete, as required in Nakai's criterion.

Definition 1.3.6 (Ampleness for $\mathbb{Q}$ -Cartier divisors). A divisor $D \in \text{Div}_{\mathbb{Q}}(X)$ is ample if the following equivalent conditions hold:

- (i) $D = \sum_i c_i A_i$, with $c_i \in \mathbb{Q}_{>0}$ and A_i ample integral Cartier divisors;

- (ii) some positive integral multiple rD has an ample integral representative;
- (iii) $(D^{\dim V} \cdot V) > 0$ for every positive-dimensional integral closed subvariety $V \subseteq X$.

The equivalence follows by clearing denominators and applying Nakai's criterion. Thus ampleness is preserved under numerical equivalence, and we can speak of ample classes in $N^1(X)_{\mathbb{Q}}$.

We now prove that ampleness is open under small perturbations.

Proposition 1.3.7. *Let X be a projective variety, let H be an ample $\mathbb{Q}$ -Cartier divisor, and let $E_1, \dots, E_r$ be $\mathbb{Q}$ -Cartier divisors. Then*

$$H + \varepsilon_1 E_1 + \dots + \varepsilon_r E_r$$

is ample for all sufficiently small rational ε_i , of either sign. In particular this holds for a single divisor E .

Proof. Clear a common denominator to assume that $H, E_1, \dots, E_r$ are integral. By Example 1.2.5, choose a fixed $m > 0$ such that all $mH \pm E_i$ are ample. If $m \sum_i |\varepsilon_i| < 1$, then

$$\begin{aligned} H + \sum_i \varepsilon_i E_i &= \left(1 - m \sum_i |\varepsilon_i|\right) H \\ &\quad + \sum_{\varepsilon_i \neq 0} m |\varepsilon_i| \left(H + \frac{\operatorname{sgn}(\varepsilon_i)}{m} E_i\right). \end{aligned}$$

Every coefficient displayed is positive, and $H \pm E_i/m = (mH \pm E_i)/m$ is ample. The expression is therefore a positive rational combination of ample divisors. For instance, it is enough to take $|\varepsilon_i| < 1/(mr)$ when $r > 0$. $\square$

Remark 1.3.8 (Weil $\mathbb{Q}$ -divisors). For an integral variety X , set

$$\operatorname{WDiv}_{\mathbb{Q}}(X) := \operatorname{WDiv}(X) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Its elements are finite rational combinations of prime divisors. The cycle map extends to

$$[\] : \operatorname{Div}_{\mathbb{Q}}(X) \longrightarrow \operatorname{WDiv}_{\mathbb{Q}}(X), \quad D \longmapsto \sum_V \operatorname{ord}_V(D)[V],$$

with the orders extended rationally from Remark 1.1.4. If X is normal, this is injective, and the orders are computed by the codimension-one valuations. In this case we identify $\operatorname{Div}_{\mathbb{Q}}(X)$ with its image in $\operatorname{WDiv}_{\mathbb{Q}}(X)$. A Weil $\mathbb{Q}$ -divisor is $\mathbb{Q}$ -Cartier if it lies in this image, equivalently if a positive multiple is an integral Cartier divisor. The operations above then apply to it.

Example 1.3.9 ($\mathbb{Q}$ -Cartier need not be Cartier). Consider the quadric cone

$$X = \{x_0x_2 - x_1^2 = 0\} \subseteq \mathbb{P}^3, \quad O = [0 : 0 : 0 : 1].$$

This surface is singular at its vertex O , but it is normal: it is a hypersurface, hence Cohen–Macaulay, and is regular in codimension one. Let $E = \{x_0 = x_1 = 0\} \subseteq X$ be a ruling line through O . It is a Weil divisor but is not Cartier at O , since its ideal (x_0, x_1) in the local ring at the vertex needs two generators. However, the tangent hyperplane $x_0 = 0$ cuts out the Cartier divisor $A = 2E$ as a Weil divisor. Hence E is $\mathbb{Q}$ -Cartier, and

$$E^2 = \left(\frac{1}{2}A \cdot \frac{1}{2}A \right) = \frac{1}{4}A^2 = \frac{1}{4} \deg X = \frac{1}{2}.$$

Thus the example illustrates the distinction between Cartier and $\mathbb{Q}$ -Cartier on a normal singular surface, not a failure of normality.

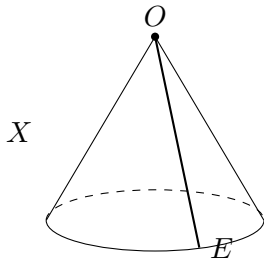


Figure 1.10: The normal quadric cone in $\mathbb{P}^3$

1.3B $\mathbb{R}$ -Cartier Divisors and Their Ampleness

Definition 1.3.10 ($\mathbb{R}$ -Cartier divisors). An $\mathbb{R}$ -Cartier divisor on a variety or scheme X is an element of

$$\mathrm{Div}_{\mathbb{R}}(X) := \mathrm{Div}(X) \otimes_{\mathbb{Z}} \mathbb{R},$$

represented by a finite sum $D = \sum_i c_i A_i$, with $c_i \in \mathbb{R}$ and $A_i \in \mathrm{Div}(X)$. Write $D_1 \sim_{\mathbb{R}} D_2$ if $D_1 - D_2$ is a finite real linear combination of principal divisors. If X is complete, intersection products extend to real-valued products and depend only on numerical equivalence. In particular,

$$D \equiv_{\mathrm{num}} 0 \iff \sum_i c_i (A_i \cdot C) = 0 \text{ for every integral curve } C \subseteq X.$$

Effectivity means that such an expression can be chosen with all $c_i \geq 0$ and all A_i effective. Supports and pullbacks are handled as above, with the same qualifications about Cartier representatives.

Example 1.3.11. There is an isomorphism

$$N^1(X)_{\mathbb{R}} \simeq N^1(X) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The key fact is linear algebra: a numerically trivial $\mathbb{R}$ -Cartier divisor is a real combination of numerically trivial integral Cartier divisors. To see it, write $B = \sum_{i=1}^s b_i B_i$. Numerical triviality is the system

$$\sum_i b_i (B_i \cdot C) = 0 \quad \text{for all integral curves } C.$$

The integer row vectors $((B_1 \cdot C), \dots, (B_s \cdot C))$ span a finite-dimensional rational vector space, so finitely many equations give the same system. Its real kernel has a rational basis; clearing denominators gives integral kernel vectors, each corresponding to a numerically trivial integral divisor. This proves the asserted fact and the isomorphism. We keep the subscript $\mathbb{R}$ when it helps distinguish the vector space from the integral group $N^1(X)$.

Definition 1.3.12 (Ampleness for $\mathbb{R}$ -Cartier divisors). Assume X is complete. An $\mathbb{R}$ -Cartier divisor is ample if it can be written as a finite positive real combination

$$D = \sum_i c_i A_i, \quad c_i > 0,$$

where the A_i are ample integral Cartier divisors. In particular, a positive real combination of ample real divisors is ample.

Remark 1.3.13 (Nakai inequalities for $\mathbb{R}$ -Cartier divisors). If D is ample, multilinearity and positivity of mixed ample intersections give

$$(D^{\dim V} \cdot V) > 0$$

for every positive-dimensional integral subvariety $V \subseteq X$. The converse is less immediate because denominators cannot be cleared. For projective varieties it is the theorem of Campana and Peternell [CP90; Laz04a, Theorem 2.3.18]. The converse also holds for complete schemes; a reference covering that generality is Fujino–Miyamoto [FM21, Theorem 1.3]. We will not use this real Nakai criterion to prove the elementary properties below.

Proposition 1.3.14 (Ample classes for $\mathbb{R}$ -Cartier divisors). *The ampleness of an $\mathbb{R}$ -Cartier divisor depends only on its numerical equivalence class.*

Proof. It is enough to prove that $D + B$ is ample if D is ample and $B \equiv_{\text{num}} 0$. By Example 1.3.11, B is a real combination of numerically trivial integral Cartier divisors. We first treat an ample integral divisor A and a numerically trivial integral divisor B_0 . For every rational r , the divisor $A + rB_0$ is ample by the

rational Nakai criterion. For arbitrary real r , choose rationals $r_1 < r < r_2$ and $0 < t < 1$ with $r = tr_1 + (1 - t)r_2$. Then

$$A + rB_0 = t(A + r_1B_0) + (1 - t)(A + r_2B_0)$$

is a positive real combination of ample rational divisors, hence is ample.

For the general case, write $D = \sum_j c_j A_j$ and $B = \sum_{i=1}^s b_i B_i$. If $s > 0$, divide the first ample summand into s parts:

$$D + B = \sum_{i=1}^s \frac{c_1}{s} \left(A_1 + \frac{sb_i}{c_1} B_i \right) + \sum_{j \geq 2} c_j A_j.$$

Every term is ample by the special case. If $B = 0$, nothing is required. $\square$

We next show that ampleness is open in the usual topology of $N^1(X)_{\mathbb{R}}$.

Example 1.3.15 (Openness of ampleness for $\mathbb{R}$ -Cartier divisors). Let X be projective, let H be an ample $\mathbb{R}$ -Cartier divisor, and let $E_1, \dots, E_r$ be $\mathbb{R}$ -Cartier divisors. Then $H + \sum_i \varepsilon_i E_i$ is ample for all sufficiently small real ε_i , of either sign.

Proof. For integral H, E_i , the exact positive-combination formula in the proof of Proposition 1.3.7 works equally well for real ε_i : the coefficients are now real and the summands remain ample rational divisors. Thus that proof already gives a real neighbourhood, not just rational points.

In general, express the finitely many E_i using finitely many integral Cartier divisors. Their new coefficients are real linear combinations of the ε_i , so it is enough to treat integral E_i . Write $H = \sum_j c_j A_j$ with $c_j > 0$ and A_j ample and integral, and choose a rational $0 < c < c_1$. Then

$$H + \sum_i \varepsilon_i E_i = c \left(A_1 + \sum_i \frac{\varepsilon_i}{c} E_i \right) + (c_1 - c)A_1 + \sum_{j \geq 2} c_j A_j.$$

For sufficiently small real ε_i , the first parenthesis is ample by the case just proved. The other nonzero summands are ample as well. $\square$

Corollary 1.3.16. *If X is projective, $N^1(X)_{\mathbb{R}}$ is spanned by classes of ample integral Cartier divisors.*

Proof. Choose integral Cartier divisors $D_1, \dots, D_\rho$ whose classes form a basis, and fix an ample Cartier divisor H . For a fixed sufficiently large m , all $mH + D_i$ and mH are ample. Since

$$[D_i] = [mH + D_i] - [mH],$$

ample integral classes span the whole space. One can consequently select a basis from this finite spanning set. The openness statement concerns the real ample cone; the integral ample classes themselves form a discrete subset, not an open subset of $N^1(X)_{\mathbb{R}}$. $\square$

1.4 Nef Line Bundles and Cartier Divisors

We have seen that if X is a projective variety, a class $\delta \in N^1(X)_{\mathbb{Q}}$ is ample if and only if it satisfies the Nakai inequalities

$$(\delta^{\dim V} \cdot V) > 0$$

for all irreducible subvarieties $V \subseteq X$ of positive dimension. So we want to make this inequality weaker:

$$(\delta^{\dim V} \cdot V) \geq 0.$$

This is related to nefness and Kleiman's theorem: for these non-negative inequalities, it suffices to test curves.

1.4A Definitions and Formal Properties

Definition 1.4.1 (Nef line bundles and Cartier divisors). Let X be a complete variety or scheme. A line bundle L on X is numerically effective, or nef, if

$$\int_C c_1(L) \geq 0$$

for every irreducible curve $C \subseteq X$. Similarly, an integral, $\mathbb{Q}$ - or $\mathbb{R}$ -Cartier divisor D is nef if $(D \cdot C) \geq 0$ for every such curve.

Since intersection with a curve depends only on numerical equivalence, this defines nef classes in $N^1(X)$, $N^1(X)_{\mathbb{Q}}$ and $N^1(X)_{\mathbb{R}}$. Any ample class is nef.

Remark 1.4.2 (Chow's lemma). In dealing with nefness, we often use Chow's lemma to reduce statements about complete varieties or schemes to the projective case. There is a projective variety or scheme X' and a proper surjective morphism

$$f : X' \longrightarrow X$$

which is an isomorphism over a dense open subset of X . On a general proper scheme the existence of an ample line bundle is not automatic. On the projective model one has ample line bundles available; nefness and intersection numbers are then compared by pullback and the projection formula.

We record the formal properties of nefness, as in [Kle66].

Example 1.4.3 (Formal properties of nefness). Let X be complete and L a line bundle on X .

- (i) If $f : Y \rightarrow X$ is proper and L is nef, then f^*L is nef. In particular, restrictions to closed subschemes remain nef.
- (ii) If, moreover, f is surjective and f^*L is nef, then L is nef.

- (iii) L is nef if and only if L_{red} is nef on X_{red} .
- (iv) L is nef if and only if its restriction to each irreducible component of X is nef.

For (i), apply the projection formula to each curve in Y , including curves contracted by f . For (ii), given a curve $C \subseteq X$, choose a curve $C' \subseteq Y$ mapping onto it with positive degree d . Such a curve can be obtained on a projective model of a component of $f^{-1}(C)$ by general hyperplane sections and then pushed down to Y . The projection formula gives $(f^*L \cdot C') = d(L \cdot C)$. The last two statements follow because the reduced irreducible curves are unchanged.

Example 1.4.4. Let X be complete and L a globally generated line bundle on X . Then L is nef.

Proof. Properness implies that $H^0(X, L)$ is finite-dimensional over $\mathbb{C}$ ([Har77, III, Remark 8.8.1]). Global generation gives a morphism

$$\varphi_{|L|} : X \longrightarrow \mathbb{P}H^0(X, L), \quad L \simeq \varphi_{|L|}^* \mathcal{O}(1).$$

By the projection formula,

$$(L \cdot C) = (\mathcal{O}(1) \cdot (\varphi_{|L|})_*[C]) \geq 0.$$

The pushforward is either zero or an effective one-cycle. □

Example 1.4.5 (Cartier divisor with nef normal bundle). Let X be complete and $D \subseteq X$ an effective Cartier divisor. If the normal bundle $N_{D/X} = \mathcal{O}_D(D)$ is nef, then D is nef.

Proof. For a curve $C \subseteq D$,

$$(D \cdot C) = \deg \mathcal{O}_C(D) = \int_C c_1(\mathcal{O}_D(D)) \geq 0.$$

For a curve not contained in D , the effective divisor restricts to an effective divisor on the normalization of C , so again $(D \cdot C) \geq 0$. □

In particular, if X is a smooth projective surface and $C \subseteq X$ is an irreducible curve with $C^2 \geq 0$, then C is nef.

1.4B Kleiman's Theorem

We discuss the fundamental result due to [Kle66].

Theorem 1.4.6 (Kleiman’s theorem). *Let X be a complete variety or scheme. If D is a nef $\mathbb{R}$ -Cartier divisor, then*

$$(D^k \cdot V) \geq 0$$

for every irreducible subvariety $V \subseteq X$ of dimension k . Similarly, $\int_V c_1(L)^{\dim V} \geq 0$ for every nef line bundle L .

Before giving a proof, we discuss some applications.

Corollary 1.4.7. *Let X be projective, D a nef $\mathbb{R}$ -Cartier divisor and H an ample $\mathbb{R}$ -Cartier divisor. Then $D + \varepsilon H$ is ample for every $\varepsilon > 0$. Conversely, if D, H are $\mathbb{R}$ -Cartier divisors and $D + \varepsilon H$ is ample for all sufficiently small $\varepsilon > 0$, then D is nef.*

Proof. “ $\Leftarrow$ ”: For every curve C ,

$$(D \cdot C) + \varepsilon(H \cdot C) > 0.$$

Letting $\varepsilon \rightarrow 0$ gives $(D \cdot C) \geq 0$.

“ $\Rightarrow$ ”: Replacing εH by H , it suffices to prove that $D + H$ is ample. We first treat the case in which $D + H$ is numerically equivalent to a rational Cartier divisor, and then make a small perturbation.

Rational case

Fix an irreducible $V \subseteq X$ of positive dimension k . Expand

$$((D + H)^k \cdot V) = \sum_{s=0}^k \binom{k}{s} (H^s D^{k-s} \cdot V).$$

Since H is a positive real combination of integral ample divisors, $H^s \cdot [V]$ has an effective real cycle representative: take sufficiently high very ample multiples and general successive hyperplane sections, and then divide by their degrees. Kleiman’s theorem applied to its components gives $(H^s D^{k-s} \cdot V) \geq 0$. The last term $(H^k \cdot V)$ is strictly positive by Example 1.2.3 and linearity. Thus $D + H$ satisfies the Nakai inequalities. Its numerical class is rational, so Nakai’s criterion and Proposition 1.3.14 imply ampleness.

General case

Choose ample integral Cartier divisors $H_1, \dots, H_r$ whose classes form a basis of $N^1(X)_{\mathbb{R}}$. By Example 1.3.15,

$$H(\varepsilon_1, \dots, \varepsilon_r) := H - \varepsilon_1 H_1 - \dots - \varepsilon_r H_r$$

remains ample for all sufficiently small $\varepsilon_i \geq 0$. For $0 < \varepsilon_i \ll 1$, the classes of $D + H(\varepsilon_1, \dots, \varepsilon_r)$ fill an open subset, and rational classes are dense. We can

therefore choose such ε_i so that $D' = D + H(\varepsilon_1, \dots, \varepsilon_r)$ has rational numerical class. The previous case makes D' ample. Hence

$$D + H = D' + \varepsilon_1 H_1 + \dots + \varepsilon_r H_r$$

is ample as a positive sum of ample real divisors. $\square$

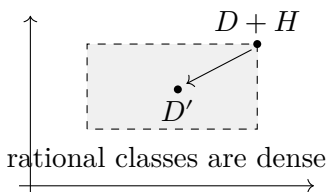


Figure 1.11: Give a small perturbation.

Corollary 1.4.8 (Test ampleness by intersecting with curves). *Let X be projective, H an ample $\mathbb{R}$ -Cartier divisor and D an $\mathbb{R}$ -Cartier divisor. Then D is ample if and only if there is an $\varepsilon > 0$ such that*

$$\frac{D \cdot C}{H \cdot C} \geq \varepsilon$$

for every irreducible curve $C \subseteq X$.

Proof. “ $\Leftarrow$ ”: Since $H \cdot C > 0$, the inequality says that $D - \varepsilon H$ is nef. Corollary 1.4.7 makes $D = (D - \varepsilon H) + \varepsilon H$ ample.

“ $\Rightarrow$ ”: If D is ample, openness of ampleness (Example 1.3.15) makes $D - \varepsilon H$ ample for $0 < \varepsilon \ll 1$. Intersecting with C gives the required inequality. $\square$

Example 1.4.9 (Comparison of ample degrees on curves). If H_1, H_2 are ample Cartier divisors on a projective variety X , there are $M, m \in \mathbb{Q}_{>0}$ such that

$$m(H_1 \cdot C) \leq (H_2 \cdot C) \leq M(H_1 \cdot C)$$

for every irreducible curve $C \subseteq X$.

Proof. By openness of ampleness, choose rational $M \gg 1$ and $0 < m \ll 1$ so that $MH_1 - H_2$ and $H_2 - mH_1$ are ample. Intersecting with C gives both bounds. $\square$

Theorem 1.4.10 (Seshadri’s criterion). *Let X be a projective variety and D a Cartier divisor. Then D is ample if and only if there is an $\varepsilon > 0$ such that*

$$\frac{D \cdot C}{\text{mult}_x C} \geq \varepsilon$$

for every closed point $x \in X$ and every irreducible curve C through x .

Proof. “ $\Rightarrow$ ”: If an effective Cartier divisor E_x passes through x and does not contain C , its local intersection number satisfies

$$(E_x \cdot C) \geq i(E_x, C; x) \geq \text{mult}_x C.$$

Choose $m > 0$ so that mD is very ample. For every pair (x, C) , a hyperplane through x not containing C gives such an $E_x \sim_{\text{lin}} mD$. Thus $(D \cdot C) \geq \frac{1}{m} \text{mult}_x C$, with the same m for all pairs.

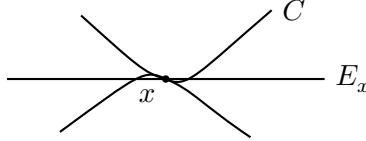


Figure 1.12: Intersection number bounded below by multiplicity.

“ $\Leftarrow$ ”: Use induction on $n = \dim X$. Dimension zero is immediate; for $n = 1$, the hypothesis gives positive degree. The same inequality holds on every proper irreducible subvariety $V \subset X$, so induction and Nakai’s criterion give $(D^{\dim V} \cdot V) > 0$. It remains to prove $D^n > 0$. Assume $n \geq 2$, fix a smooth point $x \in X$, and consider

$$\mu : X' = \text{Bl}_x(X) \longrightarrow X, \quad E = \mu^{-1}(x) \simeq \mathbb{P}^{n-1}.$$

Here $\mathcal{O}_E(E) = \mathcal{O}_{\mathbb{P}^{n-1}}(-1)$.

Step 1: We claim that $\mu^*D - \varepsilon E$ is nef on X' . If so, Kleiman’s theorem gives

$$\begin{aligned} 0 &\leq ((\mu^*D - \varepsilon E)^n)_{X'} \\ &= \sum_{s=0}^n \binom{n}{s} (-\varepsilon)^s ((\mu^*D)^{n-s} E^s)_{X'} \\ &= (D^n)_X - \varepsilon^n. \end{aligned}$$

Indeed, $\mu^*D|_E$ is trivial, so the mixed terms vanish for $1 \leq s \leq n-1$, whereas

$$(E^n)_{X'} = \int_E c_1(\mathcal{O}_E(-1))^{n-1} = (-1)^{n-1}.$$

The subscript indicates where the intersection number is computed. Consequently $D^n \geq \varepsilon^n > 0$.

Step 2: To check nefness, take an irreducible curve $C' \subseteq X'$. If C' is not contained in E , it is the strict transform of $C = \mu(C')$, and

$$(E \cdot C') = \text{mult}_x C, \quad (\mu^*D \cdot C') = (D \cdot C).$$

The multiplicity is taken to be zero if $x \notin C$. Thus

$$((\mu^*D - \varepsilon E) \cdot C') = (D \cdot C) - \varepsilon \text{mult}_x C \geq 0.$$

If $x \notin C$, use $D \cdot C > 0$, obtained by applying the hypothesis at any closed point of C . If $C' \subseteq E$, then

$$(\mu^* D \cdot C') = 0, \quad -\varepsilon(E \cdot C') = \varepsilon \deg \mathcal{O}_E(1)|_{C'} > 0.$$

This proves the claim. $\square$

Remark 1.4.11 ($(C' \cdot E) = \text{mult}_x C$). In the preceding blow-up, the strict transform C' is the blow-up of C at the ideal $\mathfrak{m}_x \mathcal{O}_C$. The exceptional divisor on this curve is $E_C = E|_{C'}$, with

$$\mathfrak{m}_x \mathcal{O}_{C'} = \mathcal{O}_{C'}(-E_C).$$

The multiplicity formula in [Ful98, p. 79], applied to the one-dimensional scheme C at x , gives $\deg E_C = \text{mult}_x C$. Therefore $(E \cdot C') = \deg E_C$. In particular, one must apply the formula to the curve here; the top self-intersection of the exceptional divisor in the ambient variety is a different intersection number.

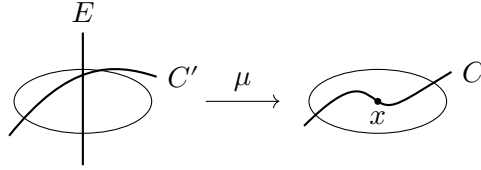


Figure 1.13: Strict transform and exceptional divisor.

Proposition 1.4.12 (Nefness in families). *Let $f : X \rightarrow T$ be a surjective proper morphism of varieties, and L a line bundle on X . Put $X_t = f^{-1}(t)$ and $L_t = L|_{X_t}$. If L_0 is nef for a given $0 \in T$, there is a countable union $B \subseteq T$ of proper closed subvarieties, not containing 0, such that L_t is nef for every $t \in T \setminus B$.*

Proof. By relative Chow's lemma there is a proper surjective modification $\mu : X' \rightarrow X$ such that $f \circ \mu : X' \rightarrow T$ is projective. Nefness of L_t is equivalent to nefness of $\mu_t^* L_t$ by Example 1.4.3. Thus we may assume f is projective. Choose a relatively ample line bundle A on X ; every A_t is ample. There is no need to represent L by a divisor avoiding every fibre. By Corollary 1.4.7,

$$L_t \text{ is nef} \iff L_t^{\otimes m} \otimes A_t \text{ is ample for every } m \geq 1.$$

For each m , ampleness in families (Theorem 1.2.11) gives an open neighbourhood U_m of 0 on which $L_t^{\otimes m} \otimes A_t$ is ample. Take $B = \bigcup_{m \geq 1} (T \setminus U_m)$, decomposing each closed subset into its finitely many irreducible components if necessary. Every such component is proper and avoids 0. For $t \notin B$, all the displayed powers are ample, and L_t is nef. $\square$

Now we turn to the proof of Kleiman's theorem.

Proof of Theorem 1.4.6. It suffices to work on each reduced irreducible subvariety and prove its top intersection non-negative. By Chow's lemma and the projection formula, we may replace this subvariety by a projective birational model. Thus assume X is projective and integral, of dimension n . We use induction on n : for $n = 1$ this is the definition of nefness, and the zero-dimensional assertion is immediate. Assume the assertion in smaller dimensions; the issue is to show $D^n \geq 0$.

Rational case

Suppose first that D is a $\mathbb{Q}$ -Cartier divisor. Fix an ample integral Cartier divisor H and consider

$$P(t) = (D + tH)^n \quad (t \in \mathbb{R}).$$

We must prove $P(0) \geq 0$. Suppose, to the contrary, that $P(0) < 0$.

Step 1: For $k < n$,

$$(D^k H^{n-k}) \geq 0.$$

Indeed, general members of a sufficiently high very ample multiple of H intersect properly to give an effective k -cycle representing $m^{n-k} H^{n-k} \cdot [X]$. Apply induction to its components and divide by m^{n-k} . Therefore every non-constant coefficient of $P(t)$ is non-negative, and the coefficient of t^n is $H^n > 0$. It follows that P is strictly increasing for $t > 0$, with a unique positive root $t_0 > 0$.

Step 2: For every rational $t > t_0$, the divisor $D + tH$ is ample. By Nakai's criterion, we check

$$((D + tH)^k \cdot V) > 0$$

for each positive-dimensional irreducible $V \subseteq X$. For $V = X$, this is $P(t) > 0$. For $V \subsetneq X$, expand

$$((D + tH)^k \cdot V) = \sum_{s=0}^k \binom{k}{s} t^{k-s} (D^s H^{k-s} \cdot V).$$

Induction, applied directly when $s = k$ and to general hyperplane sections when $s < k$, makes every term non-negative. The term $t^k (H^k \cdot V)$ is positive. This proves the claim.

Step 3: Write $P(t) = Q(t) + R(t)$, where

$$Q(t) = D \cdot (D + tH)^{n-1}, \quad R(t) = tH \cdot (D + tH)^{n-1}.$$

For rational $t > t_0$, the ample rational divisor $D + tH$ has an effective rational complete-intersection one-cycle. Its intersection with the nef D is non-negative, so $Q(t) \geq 0$. By continuity, $Q(t_0) \geq 0$. On the other hand,

$$R(t) = \sum_{s=0}^{n-1} \binom{n-1}{s} t^{n-s} (D^s H^{n-s}).$$

All coefficients are non-negative and the coefficient H^n of t^n is positive. Since $t_0 > 0$, we get $R(t_0) > 0$, contradicting $P(t_0) = Q(t_0) + R(t_0) = 0$.

General case

For a nef real divisor D , choose ample integral divisors $H_1, \dots, H_r$ whose classes form a basis of $N^1(X)_{\mathbb{R}}$. The divisors

$$D(\varepsilon_1, \dots, \varepsilon_r) = D + \varepsilon_1 H_1 + \dots + \varepsilon_r H_r$$

are nef for $\varepsilon_i > 0$, simply because their degrees on curves are non-negative. Rational numerical classes are dense in these open sets, so we may take such classes with every $\varepsilon_i \rightarrow 0$. The rational case gives $D(\varepsilon_1, \dots, \varepsilon_r)^n \geq 0$. Continuity of intersection products yields $D^n \geq 0$. The same proof applies to line bundles, using their first Chern classes on the projective models. $\square$

Example 1.4.13 (Intersection products of nef classes). Let X be a complete variety or scheme of dimension n , and let $\delta_1, \dots, \delta_n$ be nef classes. Then

$$\int_X \delta_1 \cdots \delta_n \geq 0.$$

Proof. Reduce to the projective case by taking the top-dimensional components, with their multiplicities, and applying Chow's lemma and the projection formula. Choose ample classes H_i . Corollary 1.4.7 makes $\delta_i + \varepsilon_i H_i$ ample for $\varepsilon_i > 0$, and Example 1.2.3 gives

$$\int_X (\delta_1 + \varepsilon_1 H_1) \cdots (\delta_n + \varepsilon_n H_n) > 0.$$

Let every $\varepsilon_i \rightarrow 0$. $\square$

Remark 1.4.14 (Mumford's counterexample). Let X be projective and L a line bundle. If $\int_C c_1(L) > 0$ for every irreducible curve C , does it follow that $\int_V c_1(L)^{\dim V} > 0$ for every positive-dimensional irreducible V ? The answer is no. This is the famous counterexample due to Mumford, which we will discuss later in the notes.

Definition 1.4.15 (Minimal surface). A smooth projective surface X is minimal if it contains no (-1) -curves, that is, no curve $E \simeq \mathbb{P}^1$ with $E^2 = -1$.

Example 1.4.16 (Minimal surfaces). Let X be a smooth projective surface with $\kappa(X) \geq 0$, so that $|mK_X| \neq \emptyset$ for some $m > 0$. Then X is minimal if and only if K_X is nef.

Proof. “ $\Rightarrow$ ”: Suppose K_X is not nef and take an irreducible curve C with $K_X \cdot C < 0$. Fix $D \in |mK_X|$ and write $D = \sum_i a_i C_i$, with $a_i > 0$. Since

$$(D \cdot C) = m(K_X \cdot C) < 0,$$

C must be a component, say $C = C_1$. Distinct curves have non-negative intersection, so

$$a_1 C^2 \leq D \cdot C < 0.$$

Adjunction now gives

$$-2 \leq 2p_a(C) - 2 = (K_X + C) \cdot C.$$

Both $K_X \cdot C$ and C^2 are negative integers. Hence they are both -1 and $p_a(C) = 0$, which implies that C is smooth and rational. Thus C is a (-1) -curve, contradicting minimality.

“ $\Leftarrow$ ”: For any (-1) -curve E , adjunction gives $K_X \cdot E = -1$. Such a curve cannot exist if K_X is nef. This is also the criterion in [Mat02, 1-1-4]. $\square$

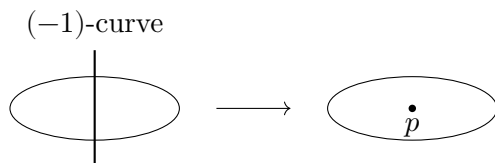


Figure 1.14: Contraction of a (-1) -curve.

Remark 1.4.17. Why do we choose the special divisor K_X here? Its role is made precise by adjunction: $K_X \cdot C + C^2 = 2p_a(C) - 2$. Together with the effective pluricanonical divisor, this forces a curve of negative canonical degree to be a (-1) -curve. Riemann–Roch is related to this formula, but the same conclusion does not hold for an arbitrary divisor in place of K_X .

1.4C Cones

Kleiman’s work [Kle66] allows us to translate positivity questions into linear algebra in a finite-dimensional real vector space V . A cone $K \subseteq V$ is a subset stable under multiplication by positive scalars; it is convex if it is also stable under addition. We do not require every cone to contain 0 , since the ample cone generally does not contain 0 . The nef cone and the cones of curves below do contain it.

Definition 1.4.18 (Ample and nef cones). The ample cone $\text{Amp}(X) \subseteq N^1(X)_{\mathbb{R}}$ consists of all ample $\mathbb{R}$ -Cartier divisor classes. The nef cone $\text{Nef}(X) \subseteq N^1(X)_{\mathbb{R}}$ consists of all nef classes. Both are convex cones.

Concretely, $\text{Amp}(X)$ consists of the finite sums $\sum_i a_i \delta_i$, where $a_i > 0$ and the δ_i are ample integral (or rational) classes.

Remark 1.4.19. As soon as $\rho(X) = \dim N^1(X)_{\mathbb{R}} \geq 3$, these cones can have quite complicated structure. For example, they need not be polyhedral.

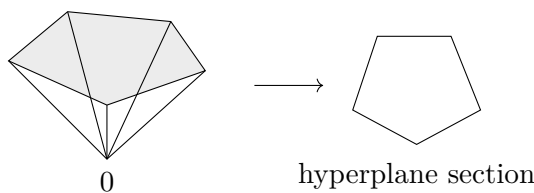


Figure 1.15: Representing cones.

Remark 1.4.20 (Visualization). It is sometimes convenient to represent a cone by its intersection with a hyperplane not passing through the origin.

We use the usual Euclidean topology on $N^1(X)_{\mathbb{R}}$ when discussing closures and interiors.

Theorem 1.4.21 (Kleiman). *Let X be a projective variety or scheme. Then*

$$(i) \text{ Nef}(X) = \overline{\text{Amp}(X)};$$

$$(ii) \text{ Amp}(X) = \text{int}(\text{Nef}(X)).$$

Proof. The nef cone is closed, since it is the intersection of the closed half-spaces defined by non-negative degree on each curve. The ample cone is open by Example 1.3.15 and lies in the nef cone. Hence

$$\overline{\text{Amp}(X)} \subseteq \text{Nef}(X), \quad \text{Amp}(X) \subseteq \text{int}(\text{Nef}(X)).$$

For the converse inclusions, fix an ample Cartier divisor H . If D is nef, Corollary 1.4.7 makes $D + \varepsilon H$ ample for every $\varepsilon > 0$. Letting $\varepsilon \rightarrow 0$ puts D in $\overline{\text{Amp}(X)}$. If $[D] \in \text{int}(\text{Nef}(X))$, then $D - \varepsilon H$ remains nef for $0 < \varepsilon \ll 1$, and

$$D = (D - \varepsilon H) + \varepsilon H$$

is ample by the same corollary. □

Remark 1.4.22 (Non-projective complete varieties). Kleiman [Kle66, pp. 325–326] proves (ii) also for a complete variety whose Zariski topology is generated by complements of effective Cartier divisors, for example a smooth or $\mathbb{Q}$ -factorial complete variety. For such an X , the nef cone has nonempty interior precisely when X is projective. The projective hypothesis cannot simply be omitted from all the assertions above without qualification.

Intersection with divisors gives a dual description in terms of curves.

Definition 1.4.23 (Numerical equivalence classes of curves). Let X be complete. The space $Z_1(X)_{\mathbb{R}}$ consists of finite real linear combinations $\gamma = \sum_i a_i [C_i]$ of irreducible curves. Two such cycles are numerically equivalent if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2) \quad \text{for every } D \in \text{Div}_{\mathbb{R}}(X).$$

The quotient is denoted by $N_1(X)_{\mathbb{R}}$. Intersection induces a perfect pairing of finite-dimensional real vector spaces

$$N^1(X)_{\mathbb{R}} \times N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \quad (\delta, \gamma) \longmapsto (\delta \cdot \gamma).$$

Numerical equivalence for integral and rational one-cycles is defined similarly. We give $N_1(X)_{\mathbb{R}}$ its Euclidean topology.

Indeed, the curve functionals span the dual of $N^1(X)_{\mathbb{R}}$: otherwise a nonzero numerical divisor class would pair to zero with every curve. This also explains finite-dimensionality and perfectness of the pairing.

Definition 1.4.24 (Cone of curves). The cone of curves is

$$\text{NE}(X) = \left\{ \sum_i a_i [C_i] : a_i \geq 0, C_i \subseteq X \text{ irreducible curves} \right\} \subseteq N_1(X)_{\mathbb{R}},$$

where the sums are finite. Its closure $\overline{\text{NE}}(X)$ is the closed cone of curves, also called the Kleiman–Mori cone.

Remark 1.4.25. The cone $\text{NE}(X)$ need not be closed.

For a cone $K \subseteq V$, its dual is

$$K^* = \{\varphi \in V^* : \varphi(v) \geq 0 \text{ for every } v \in K\}.$$

We use the following elementary convexity fact ([BV04, p. 53]).

Lemma 1.4.26. *If K is a closed convex cone containing 0 in a finite-dimensional real vector space, then $K^{**} = K$ under $V^{**} = V$.*

Proposition 1.4.27. *The closed cone of curves is dual to the nef cone:*

$$\overline{\text{NE}}(X) = \{\gamma \in N_1(X)_{\mathbb{R}} : (\delta \cdot \gamma) \geq 0 \text{ for all } \delta \in \text{Nef}(X)\}.$$

Proof. Take $V = N_1(X)_{\mathbb{R}}$ and $K = \overline{\text{NE}}(X)$. By definition and continuity, $K^* = \text{Nef}(X)$. The preceding lemma gives $K = \text{Nef}(X)^*$. $\square$

For a nonzero numerical class $[D] \in N^1(X)_{\mathbb{R}}$, write

$$\varphi_D : N_1(X)_{\mathbb{R}} \longrightarrow \mathbb{R}, \quad \gamma \longmapsto (D \cdot \gamma),$$

and set

$$D^{\perp} = \{\gamma : D \cdot \gamma = 0\}, \quad D_{>0} = \{\gamma : D \cdot \gamma > 0\}.$$

These are a hyperplane and an open half-space, respectively. The notations $D_{\geq 0}, D_{\leq 0}, D_{< 0}$ are analogous.

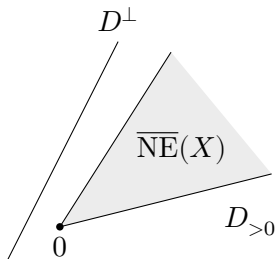


Figure 1.16: Test for ampleness via the closed cone of curves.

Theorem 1.4.28 (Kleiman’s criterion: ampleness via cones). *Let X be projective and D an $\mathbb{R}$ -Cartier divisor. Then D is ample if and only if*

$$\overline{NE}(X) \setminus \{0\} \subseteq D_{>0}.$$

Moreover, for an ample Cartier divisor H and an integer k , the set

$$W_k = \{\gamma \in \overline{NE}(X) : (H \cdot \gamma) \leq k\}$$

is compact. In particular, there are only finitely many numerical classes of effective integral one-cycles Z with $(H \cdot Z) \leq k$.

Proof. For the first assertion, we follow [Deb01].

“ $\Rightarrow$ ”: If D is ample, its degree is non-negative on $\overline{NE}(X)$ by continuity. Suppose $D \cdot \gamma = 0$ for a nonzero $\gamma \in \overline{NE}(X)$. By the perfect pairing there is a real divisor E with $E \cdot \gamma < 0$. Then $(D + tE) \cdot \gamma < 0$ for every $t > 0$, so $D + tE$ is not nef. This contradicts openness of ampleness for small t .

“ $\Leftarrow$ ”: Choose a norm on $N_1(X)_{\mathbb{R}}$. The set

$$S = \{\gamma \in \overline{NE}(X) : \|\gamma\| = 1\}$$

is compact, since it is a closed subset of the unit sphere. If X has positive dimension, S is nonempty. Positivity gives a lower bound $D \cdot \gamma \geq a > 0$ on S . For a fixed ample H , choose an upper bound $H \cdot \gamma \leq b$ with $b > 0$. Thus

$$(D - \frac{a}{b}H) \cdot \gamma \geq 0 \quad (\gamma \in S),$$

and by scaling the same holds on $\overline{NE}(X)$. Hence $D - \frac{a}{b}H$ is nef, and $D = (D - \frac{a}{b}H) + \frac{a}{b}H$ is ample by Corollary 1.4.7. In dimension zero the criterion is vacuous and every line bundle is ample.

For compactness, we keep the argument of [KM98, 1.19]. The set W_k is closed, and is empty if $k < 0$. If it were unbounded for $k \geq 0$, take $\gamma_i \in W_k$ with $\|\gamma_i\| \rightarrow \infty$. After a subsequence,

$$\frac{\gamma_i}{\|\gamma_i\|} \rightarrow \gamma \in \overline{NE}(X), \quad \|\gamma\| = 1.$$

But

$$0 \leq H \cdot \gamma = \lim_i \frac{H \cdot \gamma_i}{\|\gamma_i\|} = 0,$$

contradicting ampleness of H . Finally choose integral Cartier divisors $D_1, \dots, D_\rho$ whose classes form a real basis of $N^1(X)_\mathbb{R}$. The linear isomorphism

$$\gamma \mapsto (D_1 \cdot \gamma, \dots, D_\rho \cdot \gamma) \quad \text{from } N_1(X)_\mathbb{R} \text{ to } \mathbb{R}^\rho$$

sends integral one-cycle classes into $\mathbb{Z}^\rho$. They therefore form a discrete subset, which has finite intersection with W_k . $\square$

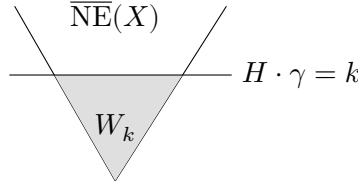


Figure 1.17: The part of $\overline{\text{NE}}(X)$ with H -degree at most k is compact.

Example 1.4.29. For a projective X , the cone $\overline{\text{NE}}(X)$ contains no line. Indeed, if $\gamma, -\gamma \in \overline{\text{NE}}(X)$ and $\gamma \neq 0$, an ample divisor would have positive intersection with both, which is impossible. More generally a line contained in this closed cone would, by rescaling and taking limits in both directions, give such γ and $-\gamma$.

Definition 1.4.30 (Extremal faces). Let K be a convex cone containing 0. An extremal face is a convex subcone $F \subseteq K$ such that

$$v, w \in K, \quad v + w \in F \quad \implies \quad v, w \in F.$$

A ray $\mathbb{R}_{\geq 0}v$ with $v \neq 0$ which is an extremal face is called an extremal ray, and is denoted by r or R .

Example 1.4.31 (Curves on a surface). On a smooth projective surface X , one-cycles and Cartier divisors are identified ([Har77, II, Proposition 6.11]). Numerical equivalence gives $N^1(X)_\mathbb{R} = N_1(X)_\mathbb{R}$, so all these cones live in the same space.

- (i) $\text{Nef}(X) \subseteq \overline{\text{NE}}(X)$, with equality if and only if $C^2 \geq 0$ for every irreducible curve $C \subseteq X$.
- (ii) If C is irreducible and $C^2 \leq 0$, then

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{C \geq 0} + \mathbb{R}_{\geq 0}[C], \quad \overline{\text{NE}}(X)_{C \geq 0} := \overline{\text{NE}}(X) \cap C_{\geq 0}.$$

- (iii) Such a $[C]$ lies on the boundary of $\overline{\text{NE}}(X)$. If $C^2 < 0$, it spans an extremal ray.

Proof. For (i), an ample integral divisor has an effective positive multiple, which on a surface is a one-cycle. Thus $\text{Amp}(X) \subseteq \overline{\text{NE}}(X)$, and taking closures gives $\text{Nef}(X) \subseteq \overline{\text{NE}}(X)$. If every $C^2 \geq 0$, then every curve is nef, since distinct curves have non-negative intersection. Closedness of the nef cone gives the reverse inclusion. Conversely, equality makes every curve class nef, hence $C^2 \geq 0$.

For (ii), first take an effective real cycle and separate its C -component: $\gamma_j = a_j[C] + \beta_j$, with $a_j \geq 0$ and β_j effective without C as a component. Then $C \cdot \beta_j \geq 0$. For a convergent sequence $\gamma_j \rightarrow \gamma \in \overline{\text{NE}}(X)$, fix ample H . The inequalities

$$0 \leq a_j(H \cdot C) \leq H \cdot \gamma_j, \quad 0 \leq H \cdot \beta_j \leq H \cdot \gamma_j$$

make a_j bounded and β_j contained in a compact slice of $\overline{\text{NE}}(X)$ by Theorem 1.4.28. Passing to a subsequence gives $\gamma = a[C] + \beta$ with $a \geq 0$ and $\beta \in \overline{\text{NE}}(X)_{C \geq 0}$.

For (iii), if $C^2 = 0$, then C is nef and the nonzero functional $\gamma \mapsto C \cdot \gamma$ vanishes at $[C]$. Its zero hyperplane supports $\overline{\text{NE}}(X)$, so $[C]$ is a boundary point. If $C^2 < 0$ and $u + v = [C]$, apply (ii) to write $u = a[C] + u_0$, $v = b[C] + v_0$, where $a, b \geq 0$ and $u_0, v_0 \in \overline{\text{NE}}(X)_{C \geq 0}$. Then

$$(1 - a - b)[C] = u_0 + v_0.$$

Intersection with C shows $1 - a - b \leq 0$; intersection with an ample H shows $1 - a - b \geq 0$. Hence $u_0 + v_0 = 0$, and positivity of H implies $u_0 = v_0 = 0$. This proves extremality. To see the boundary assertion directly, the divisor

$$H + \frac{H \cdot C}{-C^2} C$$

is nonzero and nef, and has zero intersection with C . It therefore defines a supporting hyperplane through $[C]$. $\square$

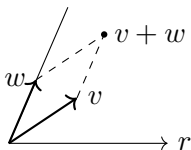


Figure 1.18: $v + w$ does not lie on the extremal ray r .

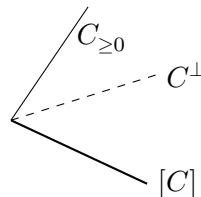


Figure 1.19: A negative curve spans an extremal ray.

1.4D Applications of Cones on Examples

To begin, we recall the simplest blow-down, following [Har77] and [Mat02]. For this result we allow any algebraically closed ground field k .

Theorem 1.4.32 (Castelnuovo's contraction theorem). *Let X be a smooth projective surface over an algebraically closed field k , and let $E \subseteq X$ be a (-1) -curve. There is a proper birational morphism*

$$\mu : X \longrightarrow X_0$$

onto a smooth projective surface, such that $\mu(E) = p$ and μ is the blow-up of X_0 at p .

Proof. We want a globally generated line bundle whose morphism contracts precisely E . In particular, it should have degree zero on E and positive degree on every other curve. Choose a very ample divisor A and put $d = A \cdot E > 0$, reserving k for the ground field. After replacing A by a sufficiently high multiple, assume $H^1(X, \mathcal{O}_X(A)) = 0$ by Serre vanishing, and redefine $d = A \cdot E$. Set

$$L = A + dE, \quad L_n = A + nE \quad (0 \leq n \leq d + 1).$$

Since $E^2 = -1$, we have $L \cdot E = 0$. For every curve $C \neq E$, $L \cdot C = A \cdot C + d(E \cdot C) > 0$.

Step 1: We claim that $H^1(X, \mathcal{O}_X(L_n)) = 0$ for $0 \leq n \leq d + 1$. The case $n = 0$ is our choice of A . For $n \geq 1$, use

$$0 \longrightarrow \mathcal{O}_X(L_{n-1}) \longrightarrow \mathcal{O}_X(L_n) \longrightarrow \mathcal{O}_E(L_n) \longrightarrow 0.$$

Since $E \simeq \mathbb{P}^1$ and $\deg \mathcal{O}_E(L_n) = d - n$, we have $\mathcal{O}_E(L_n) \simeq \mathcal{O}_{\mathbb{P}^1}(d - n)$. For $n \leq d + 1$ its first cohomology vanishes. The long exact sequence and induction prove the claim.

Step 2: We claim that $|L_n|$ is base point free for $0 \leq n \leq d$. The subsystem $|A| + nE$ shows that all possible base points lie on E . For $n \geq 1$, Step 1 makes

$$H^0(X, \mathcal{O}_X(L_n)) \longrightarrow H^0(E, \mathcal{O}_E(d - n))$$

surjective. Since $d - n \geq 0$, the bundle $\mathcal{O}_E(d - n)$ is globally generated. At each point of E , choose a section nonzero at that point and lift it. Thus there are no base points on E either. The case $n = 0$ is immediate from very ampleness of A .

Step 3: Consider

$$\varphi = \varphi_{|A+dE|} : X \longrightarrow X_1 \subseteq \mathbb{P}^N.$$

The equality $L \cdot E = 0$ implies that $\varphi(E) = p_1$ is a point: otherwise the projection formula would give positive degree on E . Away from E , the sections from $|A| + dE$ recover the embedding defined by $|A|$, since the canonical section of dE is invertible there. The projection onto these coordinates therefore gives an inverse to φ on its image away from p_1 . These coordinates all vanish on E and never vanish simultaneously away from E . Hence

$$X \setminus E \simeq X_1 \setminus \{p_1\}, \quad \varphi^{-1}(p_1) = E \quad \text{set-theoretically.}$$

Step 4: Normalize X_1 :

$$\nu : X_0 \longrightarrow X_1.$$

Since X is normal and φ is dominant, it factors as $\varphi = \nu \circ f$ with $f : X \rightarrow X_0$ proper and birational. The surface X_0 is projective, since ν is finite and X_1 is projective. The morphism f is an isomorphism away from E and maps E to a single point p over p_1 . Indeed, $\nu^{-1}(p_1)$ is finite, f is surjective, and E is connected. By normality of X_0 ,

$$f_*\mathcal{O}_X = \mathcal{O}_{X_0}$$

([Har77, III, Corollary 11.4]).

Step 5: We check smoothness at p using formal functions. Put $R = \mathcal{O}_{X_0,p}$, $\mathfrak{m} = \mathfrak{m}_p$ and $I = \mathcal{O}_X(-E)$ near $f^{-1}(p)$. At this stage we know only

$$\sqrt{\mathfrak{m}\mathcal{O}_X} = I, \quad \mathfrak{m}\mathcal{O}_X \subseteq I.$$

The two ideals have the same radical on a noetherian neighbourhood, so their powers are cofinal: for some $r > 0$, $I^r \subseteq \mathfrak{m}\mathcal{O}_X \subseteq I$. The theorem on formal functions [Har77, III, Theorem 11.1] consequently gives

$$\widehat{R} \simeq \varprojlim_n B_n, \quad B_n := H^0(E_n, \mathcal{O}_{E_n}), \quad E_n = V(I^n).$$

We have not yet asserted $\mathfrak{m}\mathcal{O}_X = I$.

Because E is Cartier with self-intersection -1 ,

$$I/I^2 \simeq \mathcal{O}_{\mathbb{P}^1}(1), \quad I^n/I^{n+1} \simeq \mathcal{O}_{\mathbb{P}^1}(n).$$

The exact sequences of infinitesimal neighbourhoods give

$$0 \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(n)) \longrightarrow B_{n+1} \longrightarrow B_n \longrightarrow 0 \quad (n \geq 1),$$

since $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$. Also $B_1 = k$. Choose compatible elements x, y in the inverse limit whose images in $\ker(B_2 \rightarrow B_1) = H^0(\mathcal{O}(1))$ form a basis. Such compatible lifts exist because every transition map is surjective.

Now keep track of multiplication. The natural maps

$$\mathrm{Sym}^n H^0(\mathcal{O}(1)) \longrightarrow H^0(\mathcal{O}(n))$$

are isomorphisms; the monomials $x^n, x^{n-1}y, \dots, y^n$ give their usual bases. Induction in the preceding exact sequences therefore gives compatible *ring* isomorphisms

$$k[x, y]/(x, y)^n \simeq B_n.$$

For $n = 1$ this is $k \simeq B_1$. In the induction step the map is an isomorphism both on the quotient B_n and on the displayed degree- n kernel, hence on B_{n+1} . Taking inverse limits yields

$$\widehat{\mathcal{O}}_{X_0,p} \simeq k[[x, y]].$$

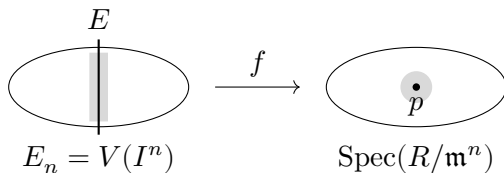


Figure 1.20: The two systems of infinitesimal neighbourhoods have cofinal ideals.

Thus $\mathcal{O}_{X_0,p}$ is a regular local ring of dimension two, and X_0 is smooth at p .

Step 6: We identify f with the blow-up at p . Choose regular parameters $u, v \in \mathfrak{m}$ whose images in $\ker(B_2 \rightarrow B_1)$ form the basis used above. Their pullbacks generate $I/I^2 \simeq \mathcal{O}_E(1)$ at every point of E , since its two standard sections generate it. Nakayama's lemma applied to $I/(\mathfrak{m}\mathcal{O}_X)$ gives

$$\mathfrak{m}\mathcal{O}_X = I = \mathcal{O}_X(-E)$$

near E ; away from E both ideals are the unit ideal. Thus the universal property of the blow-up gives a factorization

$$X \xrightarrow{u_0} \mathrm{Bl}_p(X_0) \rightarrow X_0.$$

On E , the morphism u_0 is defined by the complete system $|\mathcal{O}_{\mathbb{P}^1}(1)|$, hence maps E isomorphically onto the exceptional curve of $\mathrm{Bl}_p(X_0)$. No other curve can be contracted by u_0 , because f contracts only E . Thus u_0 is proper and quasi-finite, hence finite. It is birational with normal target, so it is an isomorphism. Set $\mu = f$. $\square$

Remark 1.4.33. The statement in [KM98] is given for smooth proper surfaces. Over an algebraically closed field this is consistent with the formulation above: Zariski's projectivity theorem says that a smooth complete algebraic surface is projective. Here the objects are schemes of finite type over the ground field, as throughout the chapter.

Corollary 1.4.34. *For every smooth projective surface, there is a finite sequence of blow-down morphisms*

$$X = X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_m$$

such that $X_i \simeq \mathrm{Bl}_{p_i}(X_{i+1})$, every X_i is smooth projective, and X_m is minimal.

Proof. If X_i contains a (-1) -curve, contract it by Castelnuovo's theorem. For a blow-up of a smooth point,

$$N^1(X_i)_{\mathbb{R}} \simeq N^1(X_{i+1})_{\mathbb{R}} \oplus \mathbb{R}[E_i],$$

so $\rho(X_{i+1}) = \rho(X_i) - 1$. Since these are non-negative integers, the process terminates. $\square$

Definition 1.4.35 (Relative cone of curves). For a morphism $\pi : X \rightarrow Y$ of projective varieties, the relative cone $\text{NE}(\pi) \subseteq \text{NE}(X)$ is generated by the classes of curves contracted by π . Here $\text{NE}(\pi)$ is not defined to include its closure.

For an ample divisor H on Y , a curve C is contracted exactly when $\pi_*[C] = 0$, equivalently $(\pi^*H \cdot C) = 0$. Since every summand of an effective real cycle has non-negative π^*H -degree,

$$\text{NE}(\pi) = \text{NE}(X) \cap \ker \pi_* = \text{NE}(X) \cap (\pi^*H)^\perp.$$

Consequently $\text{NE}(\pi)$ is closed in $\text{NE}(X)$; this sentence does not assert that it is closed in $N_1(X)_{\mathbb{R}}$.

We need the following rigidity statements; compare [Deb01, 1.15] and [KM98, 1.6]. We return to the chapter's ground field $\mathbb{C}$.

Lemma 1.4.36 (Rigidity lemma I). *Let X, Y, Y' be varieties, let $\pi : X \rightarrow Y$ be proper and surjective, and let $\pi' : X \rightarrow Y'$ be a morphism. Assume that the natural map $\mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$ is an isomorphism; in particular, every fibre of π is connected.*

(a) *If π' contracts the fibre over y_0 , there is an open neighbourhood Y_0 of y_0 and a factorization*

$$\pi'|_{\pi^{-1}(Y_0)} : \pi^{-1}(Y_0) \xrightarrow{\pi} Y_0 \longrightarrow Y'.$$

(b) *If π' contracts every fibre, then $\pi' = h \circ \pi$ for a unique morphism $h : Y \rightarrow Y'$.*

Proof. Properness of π' is unnecessary. Consider

$$g = (\pi, \pi') : X \longrightarrow Y \times Y'.$$

It is proper, since it factors as the closed graph embedding $X \rightarrow X \times Y'$ followed by the proper map $\pi \times \text{id}$. Give its closed image Z the reduced structure, and write

$$X \xrightarrow{g} Z \xrightarrow{p} Y, \quad p' : Z \longrightarrow Y', \quad \pi = p \circ g, \quad \pi' = p' \circ g.$$

The projection p is proper. Indeed, after every base change, g remains surjective; the image under p of a closed subset is the image under the proper π of its closed inverse image under g . Thus p is universally closed, and it is also separated and of finite type.

For (a), $p^{-1}(y_0) = g(\pi^{-1}(y_0))$ is a single point. A proper morphism with finite fibre is finite over some neighbourhood Y_0 of y_0 (Stacks, Tag 02OH). Set $X_0 = \pi^{-1}(Y_0)$ and $Z_0 = p^{-1}(Y_0)$. Dominance between varieties gives injections

$$\mathcal{O}_{Y_0} \subseteq (p|_{Z_0})_*\mathcal{O}_{Z_0} \subseteq (\pi|_{X_0})_*\mathcal{O}_{X_0} = \mathcal{O}_{Y_0}.$$

Hence $(p|_{Z_0})_*\mathcal{O}_{Z_0} = \mathcal{O}_{Y_0}$. Since $p|_{Z_0}$ is finite, it is an isomorphism. The desired factor is $p'|_{Z_0} \circ (p|_{Z_0})^{-1}$.

For (b), every fibre of p is finite, so p is finite everywhere and the same argument gives $Z \simeq Y$. Uniqueness follows from surjectivity of π (and the reduced structures). $\square$

Lemma 1.4.37 (Rigidity lemma II). *Let Y be an irreducible variety and $f : Y \rightarrow Z$ a proper surjective morphism whose fibres are connected and all have dimension n . If a morphism $g : Y \rightarrow X$ contracts one fibre $f^{-1}(z_0)$, then it contracts every fibre of f .*

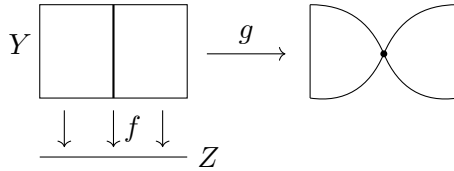
Proof. As above, $(f, g) : Y \rightarrow Z \times X$ is proper. Let W be its reduced closed image, and factor

$$f : Y \xrightarrow{h} W \xrightarrow{p} Z.$$

Both maps are proper and surjective. Properness of p follows by the universally closed argument in the preceding proof; surjectivity alone would not allow us to lift a prescribed K -point without extending K . We have $p^{-1}(z) = h(f^{-1}(z))$ and $p^{-1}(z_0)$ is a point.

Upper semicontinuity of fibre dimension gives a nonempty open $U \subseteq Z$ such that every fibre of p over U is finite. Since $f^{-1}(z)$ is connected, this finite image is one point. Thus the fibres of h over $p^{-1}(U)$ have dimension n . The irreducibility of W and upper semicontinuity for the proper h show that every fibre of h has dimension at least n .

For $w \in p^{-1}(z)$, we have $h^{-1}(w) \subseteq f^{-1}(z)$ and $\dim f^{-1}(z) = n$. Hence $h^{-1}(w)$ has dimension n and contains an n -dimensional irreducible component of $f^{-1}(z)$. Distinct w give disjoint fibres, and there are only finitely many such components. Therefore $p^{-1}(z)$ has only finitely many points. It is connected as the image of the connected $f^{-1}(z)$, so it is a single point. This says exactly that g contracts $f^{-1}(z)$. $\square$



Only one fibre contracted: impossible under Lemma 1.4.37.

Proposition 1.4.38. *Let X, Y, Y' be projective varieties and $\pi : X \rightarrow Y$ a morphism.*

- (a) *The subcone $\text{NE}(\pi)$ is an extremal face of $\text{NE}(X)$.*
- (b) *Assume $\pi_* \mathcal{O}_X = \mathcal{O}_Y$, and let $\pi' : X \rightarrow Y'$ be a morphism. If $\text{NE}(\pi) \subseteq \text{NE}(\pi')$, there is a unique morphism $h : Y \rightarrow Y'$ with $\pi' = h \circ \pi$. Consequently, among morphisms satisfying $\pi_* \mathcal{O}_X = \mathcal{O}_Y$, the morphism π is determined up to isomorphism by $\text{NE}(\pi)$.*

Proof. For (a), take $a = \sum_i a_i [C_i]$ and $a' = \sum_j a'_j [C'_j]$ in $\text{NE}(X)$, with all coefficients positive after deleting zero terms. If $a + a' \in \text{NE}(\pi)$, pushing forward gives

$$\sum_i a_i \pi_* [C_i] + \sum_j a'_j \pi_* [C'_j] = 0 \quad \text{in } N_1(Y)_{\mathbb{R}}.$$

Intersect with an ample divisor H on Y . Each term is non-negative, and is positive unless the corresponding curve is contracted. Hence all these curves are contracted, so $a, a' \in \text{NE}(\pi)$. The point is that no nonzero effective real one-cycle on a projective variety is numerically trivial.

For (b), each curve contracted by π is contracted by π' . On each projective irreducible component of a fibre of π , a morphism with positive-dimensional image would map some curve onto a curve (by taking general sections over a curve in the image). Thus π' maps every such component to a point. Connectedness of the fibre makes all these points equal. Rigidity lemma I(b) gives the unique factor h . If two morphisms satisfying the stated pushforward condition have the same relative cone, apply the factorization in both directions. Surjectivity makes the two factors inverse isomorphisms. $\square$

Remark 1.4.39. An ample divisor H on the projective target Y has positive degree on every irreducible curve. If $C \subseteq X$ is not contracted, then

$$\pi_* [C] = \deg(C \rightarrow \pi(C)) [\pi(C)]$$

has positive H -degree. Thus $\pi_* [C] = 0$ numerically forces C to be contracted. This does not mean that distinct curves on Y cannot be numerically equivalent.

Example 1.4.40 (Picard number one). If X is projective and $\dim N_1(X)_{\mathbb{R}} = 1$, for example $X = \mathbb{P}^n$ with $n \geq 1$, then $\text{NE}(X) = \overline{\text{NE}}(X)$ is a single ray. Its only extremal faces are $\{0\}$ and $\text{NE}(X)$ itself. They correspond to the identity $X \rightarrow X$ and the map to a point, respectively.

$$\begin{array}{c} \text{NE}(X) \\ \xrightarrow{\quad} N_1(X)_{\mathbb{R}} \\ \bullet \\ 0 \end{array}$$

Example 1.4.41 (Product of projective spaces). Let $X = \mathbb{P} \times \mathbb{P}'$ be the product of two positive-dimensional projective spaces. Then $N_1(X)_{\mathbb{R}}$ has dimension two, spanned by the class l of a line in $\mathbb{P} \times \{p'\}$ and the class l' of a line in $\{p\} \times \mathbb{P}'$. They are dual to the two pulled-back hyperplane classes. The cone is

$$\text{NE}(X) = \overline{\text{NE}}(X) = \mathbb{R}_{\geq 0} l + \mathbb{R}_{\geq 0} l'.$$

Its two extremal rays are generated by l and l' . They correspond, respectively, to the projections $X \rightarrow \mathbb{P}'$ and $X \rightarrow \mathbb{P}$. Indeed, the projection to $\mathbb{P}'$ contracts the lines lying in the first factor, and the projection to $\mathbb{P}$ contracts those in the second.

1.4E Fujita's Vanishing Theorem

Fujita's theorem makes Serre-type vanishing uniform with respect to twists by nef Cartier divisors.

Theorem 1.4.42 (Fujita's vanishing theorem). *Let X be a complex projective scheme and H an ample integral Cartier divisor. For every coherent sheaf $\mathcal{F}$ there is an integer $m(\mathcal{F}, H)$ such that*

$$H^i(X, \mathcal{F} \otimes \mathcal{O}_X(mH + D)) = 0$$

for every $i > 0$, every $m \geq m(\mathcal{F}, H)$ and every nef integral Cartier divisor D .

The essential point is that $m(\mathcal{F}, H)$ is independent of D . Applying Serre vanishing separately for each D would not give this uniformity. The proof below uses characteristic-zero vanishing for a resolution; that vanishing theorem is an input independent of Fujita's theorem.

Proof. Replacing X by the closed subscheme defined by the annihilator of $\mathcal{F}$, we may assume $\text{Supp } \mathcal{F} = X$ set-theoretically. We use induction on dimension, allowing every coherent sheaf in the induction hypothesis.

Step 1: In dimension zero, all higher cohomology vanishes. Henceforth assume the assertion known in smaller dimensions.

Step 2: Reduce to the case in which X is reduced. Let N be the nilradical of $\mathcal{O}_X$, with $N^r = 0$. Use the filtration

$$\mathcal{F} \supseteq N\mathcal{F} \supseteq N^2\mathcal{F} \supseteq \dots \supseteq N^r\mathcal{F} = 0.$$

Every quotient $Q_i = N^i\mathcal{F}/N^{i+1}\mathcal{F}$ is annihilated by N , hence is a coherent $\mathcal{O}_{X_{\text{red}}}$ -module. If the theorem holds on reduced schemes, it applies to each Q_i , since $H|_{X_{\text{red}}}$ is ample and $D|_{X_{\text{red}}}$ is nef. Take the maximum of the finitely many bounds. The sequences

$$0 \rightarrow N^{i+1}\mathcal{F} \rightarrow N^i\mathcal{F} \rightarrow Q_i \rightarrow 0,$$

twisted by $\mathcal{O}_X(mH + D)$, prove vanishing for $N^i\mathcal{F}$ by decreasing induction on i , and finally for $\mathcal{F}$.

Step 3: Reduce from reduced X to integral X . Write $X = X_1 \cup \dots \cup X_s$ with its reduced irreducible components. Let I be the ideal of X_1 and J the ideal of $X_2 \cup \dots \cup X_s$, with reduced structure. Then $IJ = 0$, because $I \cap J = 0$ on the reduced union. In

$$0 \rightarrow I\mathcal{F} \rightarrow \mathcal{F} \rightarrow \mathcal{F}/I\mathcal{F} \rightarrow 0,$$

the first term is annihilated by J and is a sheaf on the other components; the quotient is a sheaf on X_1 . Induction on the number of components, followed by the long exact cohomology sequence, proves the assertion using the maximum of the two bounds.

Step 4: We may assume H very ample. Choose $\ell > 0$ such that ℓH is very ample. If the assertion holds with ℓH , apply it to $\mathcal{F}(rH)$ for $0 \leq r < \ell$. Choose a common non-negative bound q_0 for these finitely many sheaves. Write $m = q\ell + r$; when $m \geq \ell q_0$, we have $q \geq q_0$, and

$$\mathcal{F}(mH + D) = \mathcal{F}(rH) \otimes \mathcal{O}_X(q\ell H + D)$$

has the required vanishing.

Step 5: It suffices to prove the assertion for $\mathcal{F} = \mathcal{O}_X$ on integral X . We induct on $\text{rank } \mathcal{F}$. Rank zero means that $\mathcal{F}$ is supported in smaller dimension, so dimension induction applies. If the rank is positive, take $a \gg 0$ so that $\mathcal{F}(aH)$ is globally generated, and choose a global section s which is *nonzero at the generic point* η . It defines

$$\alpha : \mathcal{O}_X \longrightarrow \mathcal{F}(aH)$$

which is injective at η . Its kernel is an ideal subsheaf of the torsion-free $\mathcal{O}_X$, so it is zero. Explicitly, on an affine open $\text{Spec } A$ the kernel is an ideal $I \subseteq A$ with $I \otimes_A K(A) = 0$; the injection $I \hookrightarrow K(A)$ forces $I = 0$. Complete this to

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\alpha} \mathcal{F}(aH) \longrightarrow \mathcal{G} \longrightarrow 0, \quad \text{rank } \mathcal{G} = \text{rank } \mathcal{F} - 1.$$

After twisting by $\mathcal{O}_X((m-a)H + D)$, the long exact sequence gives the claim for $\mathcal{F}$ from the claims for $\mathcal{O}_X$ and $\mathcal{G}$. It is enough to take

$$m(\mathcal{F}, H) \geq a + \max\{m(\mathcal{O}_X, H), m(\mathcal{G}, H)\}.$$

Step 6: Let $f : Y \rightarrow X$ be a projective resolution of singularities. The coherent sheaf $f_*\omega_Y$ is torsion-free of rank one. For $b \gg 0$, its dual twisted by bH is globally generated. Choose a section nonzero at the generic point to obtain an injection

$$f_*\omega_Y \longrightarrow \mathcal{O}_X(bH).$$

The kernel is zero because it is generically zero and is contained in a torsion-free sheaf. The cokernel $\mathcal{Q}$ has smaller-dimensional support, giving

$$0 \longrightarrow f_*\omega_Y \longrightarrow \mathcal{O}_X(bH) \longrightarrow \mathcal{Q} \longrightarrow 0.$$

For $m > 0$ and D nef, $mH + D$ is ample by Corollary 1.4.7. Kollár's vanishing theorem [Laz04a, Theorem 4.3.8] gives

$$H^i(X, f_*\omega_Y \otimes \mathcal{O}_X(mH + D)) = 0 \quad (i > 0).$$

The dimension induction hypothesis applies to $\mathcal{Q}$, uniformly in D . Thus, for $m \geq \max\{1, m(\mathcal{Q}, H)\}$,

$$H^i(X, \mathcal{O}_X((b+m)H + D)) = 0 \quad (i > 0).$$

This proves the case $\mathcal{F} = \mathcal{O}_X$ and completes all the reductions. □

Remark 1.4.43. Fujita vanishing also holds over algebraically closed fields of positive characteristic; see [Fuj11]. The resolution-and-vanishing proof above is a characteristic-zero proof, so a different argument is needed there.

Theorem 1.4.44 (Higher cohomology of nef Cartier divisors, II). *Let X be a projective variety or scheme of dimension n , and let D be a nef Cartier divisor. For every coherent sheaf $\mathcal{F}$ and every $0 \leq i \leq n$,*

$$h^i(X, \mathcal{F}(mD)) = O(m^{n-i}) \quad (m \rightarrow +\infty).$$

For $i > n$ the cohomology is zero.

Proof. The case $i = 0$ follows from Example 1.2.24, without using nefness. For higher cohomology, induct on n , with dimension zero immediate. By Fujita vanishing, choose a fixed very ample divisor class H such that

$$H^i(X, \mathcal{F}(mD + H)) = 0 \quad (i > 0, m \geq 0).$$

Here the variable m multiplies the nef divisor; it is the fixed ample multiple used to choose H that is made sufficiently large. Choose a general member of $|H|$ avoiding every associated subvariety of $\mathcal{F}$ and of $\mathcal{O}_X$, and denote this effective Cartier divisor again by H . Multiplication by its section is injective on $\mathcal{F}$, so

$$0 \longrightarrow \mathcal{F}(mD) \longrightarrow \mathcal{F}(mD + H) \longrightarrow \mathcal{F}(mD + H) \otimes \mathcal{O}_H \longrightarrow 0.$$

For $i \geq 1$, the long exact sequence gives

$$\begin{aligned} h^i(X, \mathcal{F}(mD)) &\leq h^{i-1}(H, (\mathcal{F}(H) \otimes \mathcal{O}_H)(mD)) \\ &= O(m^{(n-1)-(i-1)}) = O(m^{n-i}) \end{aligned}$$

by induction, since $D|_H$ is nef. The cohomology groups are isomorphic for $i \geq 2$; for $i = 1$ one has only a surjection from the group on H . No lower bound on the nef exponent m is needed for the chosen vanishing. $\square$

Remark 1.4.45. The associated-prime condition is exactly what ensures injectivity. On an affine open, write $\mathcal{F} = \widetilde{M}$ and let h be a local equation for H . After trivializing the line-bundle factors, the map $\mathcal{F}(mD) \rightarrow \mathcal{F}(mD + H)$ is multiplication by h on M . It is injective if and only if h is not a zero divisor on M , equivalently

$$h \notin \bigcup_{\mathfrak{p} \in \text{Ass } M} \mathfrak{p}.$$

If h lies in an associated prime, there is some nonzero element of M annihilated by h ; this need not hold for every element. This is why we choose the general divisor to avoid the associated subvarieties.

Corollary 1.4.46 (Asymptotic Riemann–Roch for nef Cartier divisors). *Let X be a projective scheme of dimension n and D a nef Cartier divisor. Then*

$$h^0(X, \mathcal{O}_X(mD)) = \frac{D^n \cdot [X]_n}{n!} m^n + O(m^{n-1}),$$

where $[X]_n$ is the top-dimensional cycle, including its multiplicities. For integral X , the usual notation is

$$h^0(X, \mathcal{F}(mD)) = \text{rank}(\mathcal{F}) \frac{D^n}{n!} m^n + O(m^{n-1}).$$

For an arbitrary projective scheme and coherent $\mathcal{F}$, the general form is

$$h^0(X, \mathcal{F}(mD)) = \frac{D^n \cdot Z_n(\mathcal{F})}{n!} m^n + O(m^{n-1}),$$

with

$$Z_n(\mathcal{F}) = \sum_{\dim V=n} \text{length}_{\mathcal{O}_{X, \eta_V}}(\mathcal{F}_{\eta_V})[V].$$

In particular, if $\mathcal{F}$ is locally free of constant rank r , then $Z_n(\mathcal{F}) = r[X]_n$, recovering $rD^n/n!$ with the usual scheme-theoretic intersection convention.

Proof. Asymptotic Riemann–Roch gives the stated leading coefficient for $\chi(X, \mathcal{F}(mD))$. Theorem 1.4.44 makes every higher-cohomology contribution $O(m^{n-1})$, so the same leading coefficient holds for h^0 . In dimension zero the formula is the exact length formula. $\square$

1.5 Ampleness in the Relative Setting

Ampleness can be a relative concept, in the spirit of Grothendieck’s relative point of view. Let $f : X \rightarrow T$ be a proper morphism of schemes and $\mathcal{F}$ a coherent sheaf on X . Our finite-type convention makes T noetherian and quasi-compact. The sheaf $f_*\mathcal{F}$ is coherent, and we can form

$$\pi : \mathbb{P}(f_*\mathcal{F}) := \text{Proj}_T \text{Sym}(f_*\mathcal{F}) \longrightarrow T.$$

Its fibre over t is the projective space of one-dimensional quotients of $(f_*\mathcal{F}) \otimes \kappa(t)$. This vector space need not equal $H^0(X_t, \mathcal{F}_t)$ unless the relevant cohomology-and-base-change statement holds. If $f_*\mathcal{F}$ is free of rank $n+1$ on $U \subseteq T$, then $\pi^{-1}(U) \simeq \mathbb{P}_U^n$. The natural evaluation map

$$f^*f_*\mathcal{F} \longrightarrow \mathcal{F}$$

is the relative analogue of global generation. For $T = \text{Spec } \mathbb{C}$, it is $H^0(X, \mathcal{F}) \otimes_{\mathbb{C}} \mathcal{O}_X \rightarrow \mathcal{F}$, as in Example 1.4.4.

1.5A Relative Ampleness

Definition 1.5.1 (Ampleness for a map). Let $f : X \rightarrow T$ be proper and L a line bundle on X .

- (i) L is f -free if $\rho : f^*f_*L \rightarrow L$ is surjective. It is f -very ample if, in addition, the resulting morphism

$$j : X \longrightarrow \mathbb{P}(f_*L)$$

is a closed immersion over T .

- (ii) L is f -ample if $L^{\otimes m}$ is f -very ample for some $m > 0$.

An integral Cartier divisor is relatively ample or very ample when its associated line bundle is. Relative ampleness for $\mathbb{Q}$ -Cartier divisors is defined by a positive multiple clearing denominators.

Remark 1.5.2 (Relative ampleness is local on the base). Both properties are local on T . For very ampleness, one can equivalently use an open cover $\{U_\alpha\}$ and closed immersions

$$j_\alpha : X_\alpha = f^{-1}(U_\alpha) \hookrightarrow \mathbb{P}_{U_\alpha}^{n_\alpha}, \quad L|_{X_\alpha} \simeq j_\alpha^* \mathcal{O}(1).$$

See [GW20, 13.56 and 13.63]. For ampleness, the exponents can be made uniform on a finite subcover, using a common positive multiple. If $U \subseteq T$ is affine, L is f -ample over U exactly when $L|_{f^{-1}(U)}$ is ample in the absolute sense.

A proper morphism with a relatively ample line bundle is projective in the convention of embedding into $\mathbb{P}(\mathcal{E})$ for a coherent sheaf $\mathcal{E}$ on T . This is the convention used here. An embedding into one fixed $\mathbb{P}_T^N$ is a stronger global formulation over a general base; locally on T it is available. Compare Stacks, Tags 01VV and 01W7.

Example 1.5.3. For a vector bundle $\mathcal{E}$ on T , the Serre line bundle $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(1)$ is relatively very ample, hence ample, for $\pi : \mathbb{P}(\mathcal{E}) \rightarrow T$.

The cohomological properties are as follows.

Theorem 1.5.4. *Let $f : X \rightarrow T$ be proper and L a line bundle. The following are equivalent:*

- (i) L is f -ample.
(ii) For every coherent $\mathcal{F}$, there is $m_1(\mathcal{F}) > 0$ such that

$$R^i f_*(\mathcal{F} \otimes L^{\otimes m}) = 0 \quad (i > 0, m \geq m_1(\mathcal{F})).$$

- (iii) For every coherent $\mathcal{F}$, there is $m_2(\mathcal{F}) > 0$ such that

$$f^* f_*(\mathcal{F} \otimes L^{\otimes m}) \longrightarrow \mathcal{F} \otimes L^{\otimes m}$$

is surjective for every $m \geq m_2(\mathcal{F})$.

(iv) There is $m_3 > 0$ such that $L^{\otimes m}$ is f -very ample for all $m \geq m_3$.

These are the relative Serre criteria; they can be checked on a finite affine cover of the noetherian base.

Example 1.5.5. Suppose $Y \xrightarrow{\mu} X \xrightarrow{f} T$, where μ is finite, and put $g = f \circ \mu$. If L is f -ample, then μ^*L is g -ample.

Proof. Take an affine cover $\{U_i\}$ of T . Then $L|_{f^{-1}(U_i)}$ is ample, and the restriction $g^{-1}(U_i) \rightarrow f^{-1}(U_i)$ of μ is finite. Finite pullback preserves ampleness, without a surjectivity assumption. Thus μ^*L is relatively ample on every U_i , hence on T . $\square$

Theorem 1.5.6 (Fibrewise ampleness). *Let $f : X \rightarrow T$ be proper and L a line bundle. For a closed point $t \in T$, put $X_t = f^{-1}(t)$ and $L_t = L|_{X_t}$. Then L is f -ample if and only if L_t is ample for every closed point $t \in T$. Equivalently, one may test every geometric fibre.*

Proof. If L is f -ample, restrict a relative embedding to a fibre. A positive power of L_t is very ample, so L_t is ample. Conversely, the last step in the proof of ampleness in families (Theorem 1.2.11) gives, near each closed t , an open neighbourhood U_t , an integer $m_t > 0$ and a finite morphism

$$\varphi_t : X_{U_t} \longrightarrow \mathbb{P}^{r_t} \times U_t$$

with

$$L^{\otimes m_t}|_{X_{U_t}} \simeq \varphi_t^* \mathcal{O}_{\mathbb{P}^{r_t} \times U_t}(1).$$

The preceding example makes this power relatively ample over U_t , and hence so is L . These open sets cover T : their complement, if nonempty, is a closed subset of a finite-type scheme over a field and contains a closed point. Relative ampleness is local on the base, so the claim follows. The formulation with geometric fibres follows by base change and descent of ampleness under field extension. $\square$

Corollary 1.5.7 (Relative Nakai–Moishezon criterion). *For a proper $f : X \rightarrow T$, a $\mathbb{Q}$ -Cartier divisor D is f -ample if and only if*

$$(D^{\dim V} \cdot V) > 0$$

for every positive-dimensional irreducible closed subvariety $V \subseteq X$ which maps to a closed point of T .

The intersection is computed on that proper fibre, using the associated line bundles. This avoids requiring the Cartier divisor itself to restrict as a Cartier divisor when its support contains V .

Proposition 1.5.8. *Let $f : X \rightarrow T$ be a morphism of projective schemes, L a line bundle on X , and A an ample line bundle on T . Then L is f -ample if and only if $L \otimes f^*A^{\otimes m}$ is ample for all $m \gg 0$. For the reverse implication, one such m already suffices.*

Proof. “ $\Rightarrow$ ”: Choose $r > 0$ such that $L^{\otimes r}$ is f -free. For $p \gg 0$, the coherent sheaf $f_*L^{\otimes r} \otimes A^{\otimes p}$ is globally generated. Pull back its generators and use the evaluation surjection to obtain an everywhere-defined morphism

$$\varphi : X \longrightarrow \mathbb{P} \times T, \quad \mathbb{P} = \mathbb{P}H^0(T, f_*L^{\otimes r} \otimes A^{\otimes p}),$$

with

$$L^{\otimes r} \otimes f^*A^{\otimes p} \simeq \varphi^* \text{pr}_1^* \mathcal{O}_{\mathbb{P}}(1).$$

This morphism is finite: it is proper, and on every fibre of f the pulled-back $\mathcal{O}(1)$ is the ample $L_t^{\otimes r}$, so a positive-dimensional fibre of φ would contain a curve of degree zero, a contradiction. For every integer $q \geq 1$,

$$L^{\otimes r} \otimes f^*A^{\otimes(p+q)} \simeq \varphi^*(\mathcal{O}_{\mathbb{P}}(1) \boxtimes A^{\otimes q})$$

is ample by finite pullback. In particular, for all $m \gg 0$, take $q = rm - p \geq 1$. Then

$$(L \otimes f^*A^{\otimes m})^{\otimes r} = L^{\otimes r} \otimes f^*A^{\otimes rm}$$

is ample, proving the assertion for the original L .

“ $\Leftarrow$ ”: Restriction to X_t gives

$$(L \otimes f^*A^{\otimes m})|_{X_t} \simeq L_t,$$

up to tensoring by the one-dimensional vector space $A(t)^{\otimes m}$. Thus each L_t is ample, and Theorem 1.5.6 applies. $\square$

1.5B Intersections in the Relative Setting

Corollary 1.5.7 characterizes relative ampleness by intersections on fibres. This motivates relative numerical equivalence. Let $f : X \rightarrow T$ be proper. As in Section 1.3, Cartier notation agrees with line-bundle notation on varieties and on schemes where line bundles have Cartier representatives. For arbitrary schemes without this property, the numerical groups and real positivity statements below are understood using numerical classes of line bundles. This gives the same formulas wherever Cartier representatives are available.

Definition 1.5.9 (Relative subvariety). A closed irreducible subvariety $Z \subseteq X$ is relative over T if $f(Z)$ is a closed point of T .

Such a Z is proper over the ground field, so its intersection numbers with line bundles on X are defined even when X itself is not proper absolutely.

Definition 1.5.10 (Relative numerical equivalence). Two Cartier divisors D_1, D_2 are relatively numerically equivalent, written $D_1 \equiv_T D_2$ or $D_1 \equiv_f D_2$, if

$$(D_1 \cdot C) = (D_2 \cdot C) \quad \text{for every relative curve } C \subseteq X.$$

The subgroup of relatively numerically trivial divisors is $\text{Num}(X/T) = \{D : D \equiv_T 0\}$. The analogous definition for line bundles uses their degrees on relative curves.

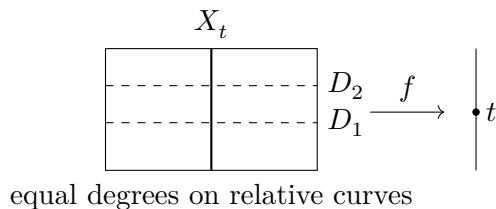


Figure 1.21: Relative numerical equivalence.

Definition 1.5.11 (Relative numerical divisor group). In Cartier notation, the relative numerical divisor group is

$$N^1(X/T) = \text{Div}(X) / \text{Num}(X/T).$$

It is also called the relative Néron–Severi group with the numerical convention adopted in these notes. On arbitrary schemes, use $\text{Pic}(X)$ modulo line bundles of degree zero on all relative curves, as specified above. Set $\rho(X/T) = \text{rank } N^1(X/T)$.

If X is complete, absolute numerical triviality implies relative numerical triviality, so there is a surjection

$$N^1(X) \longrightarrow N^1(X/T), \quad N^1(X/T) \simeq \frac{\text{Div}(X) / \text{Num}(X)}{\text{Num}(X/T) / \text{Num}(X)}$$

in Cartier notation. This proves finite generation in this case. For non-proper X , one cannot appeal to the absolute group as previously defined only for complete schemes.

For completeness, over $\mathbb{C}$ one can see the general finiteness as follows. By Chow’s lemma and resolution, take a proper surjective morphism $\mu : Y \rightarrow X$ with Y a finite disjoint union of smooth quasi-projective varieties. Choose smooth projective compactifications $\bar{Y}$. Every line bundle on Y extends to $\bar{Y}$ by taking the closure of a Cartier divisor on the smooth compactification. Thus the finite numerical groups of $\bar{Y}$ surject onto the numerical groups on Y tested against curves relative to T . Pullback from X is injective on these numerical groups: a relative curve on X has a curve above it mapping onto it with positive degree, and the projection formula detects its degree. Hence $N^1(X/T)$ is finitely generated and torsion-free. In particular, $\rho(X/T)$ is finite.

Definition 1.5.12 (Real coefficients). For real Cartier divisors, write $D_1 \equiv_T D_2$ if their degrees on every relative curve agree, with intersections extended real-linearly. The resulting vector space is

$$N^1(X/T)_{\mathbb{R}} = N^1(X/T) \otimes_{\mathbb{Z}} \mathbb{R}.$$

The corresponding definition applies to real linear combinations of line bundles under the general scheme convention.

The identification follows as in Example 1.3.11: the relative curve conditions are homogeneous linear equations with integral coefficients.

Definition 1.5.13 (Numerical classes of relative curves). Let $f : X \rightarrow T$ be proper. The space $Z_1(X/T)_{\mathbb{R}}$ consists of finite real sums $\gamma = \sum_i a_i [C_i]$ of relative irreducible curves. Two such cycles are numerically equivalent if

$$(D \cdot \gamma_1) = (D \cdot \gamma_2)$$

for every Cartier divisor D on X (or every line bundle, with the general scheme convention). Denote the quotient by $N_1(X/T)_{\mathbb{R}}$. There is a perfect pairing

$$N^1(X/T)_{\mathbb{R}} \times N_1(X/T)_{\mathbb{R}} \longrightarrow \mathbb{R}.$$

In particular both spaces are finite-dimensional, and we give them their Euclidean topology. Integral and rational cycle classes are defined similarly.

For a proper morphism $g : Y \rightarrow X$ over T and a relative integral subvariety $Z \subseteq Y$, proper pushforward is

$$g_*[Z] = \begin{cases} 0, & \dim g(Z) < \dim Z, \\ d[g(Z)], & \dim g(Z) = \dim Z, \end{cases} \quad d = [k(Z) : k(g(Z))].$$

The target cycle lies on X , and the projection formula holds.

Proposition 1.5.14. *Let $f : X \rightarrow T$ and $g : Y \rightarrow X$ be proper.*

(1) *If $L \equiv_T 0$, then $g^*L \equiv_T 0$. Thus pullback induces a group homomorphism*

$$g^* : N^1(X/T) \longrightarrow N^1(Y/T),$$

and an $\mathbb{R}$ -linear map after tensoring with $\mathbb{R}$.

(2) *If g is surjective, this map is injective.*

Proof. For (1), take a relative curve $C \subseteq Y$. If g contracts it, $(g^*L \cdot C) = 0$. Otherwise

$$(g^*L \cdot C) = (L \cdot g_*[C]) = d(L \cdot g(C)) = 0.$$

For (2), every relative curve $C \subseteq X$ is dominated by a relative curve $C' \subseteq Y$ of positive degree d , as in Example 1.4.3(ii). Thus

$$(L \cdot C) = \frac{1}{d}(g^*L \cdot C'),$$

which proves the assertion. For $T = \text{Spec } \mathbb{C}$ this is the absolute statement. $\square$

Definition 1.5.15. A real Cartier divisor is f -ample if it is a finite positive real combination of f -ample integral Cartier divisors. The rational definition is analogous and agrees with clearing denominators. For real line-bundle classes, make the same definition in $\text{Pic}(X) \otimes \mathbb{R}$; this is the interpretation in general scheme statements. When T is a point, these are the absolute definitions.

Proposition 1.5.16 (Openness of relative ampleness). *Let $f : X \rightarrow T$ be a projective morphism, and let H be an f -ample real divisor (or real line-bundle class). Given finitely many real divisors $E_1, \dots, E_r$, the class*

$$H + \varepsilon_1 E_1 + \dots + \varepsilon_r E_r$$

is f -ample for all sufficiently small real ε_i .

Proof. Step 1: First assume $H, E_1, \dots, E_r$ integral. Relative Serre ampleness, checked on a finite affine cover of T , makes all the $2r$ divisors $mH \pm E_i$ relatively ample for one sufficiently large m . This is the relative form of Example 1.2.5. When X, T are projective, we can also use the original absolute argument: choose ample A on T and $n \gg 0$ so that $H + nf^*A$ is ample by Proposition 1.5.8. Then

$$m(H + nf^*A) \pm E_i = (mH \pm E_i) + mnf^*A$$

is ample for $m \gg 0$, and restriction to fibres makes $mH \pm E_i$ relatively ample. For any real ε_i with $m \sum_i |\varepsilon_i| < 1$, use the exact decomposition

$$\begin{aligned} H + \sum_i \varepsilon_i E_i &= \left(1 - m \sum_i |\varepsilon_i|\right) H \\ &\quad + \sum_{\varepsilon_i \neq 0} m|\varepsilon_i| \left(H + \frac{\text{sgn}(\varepsilon_i)}{m} E_i\right). \end{aligned}$$

Every divisor in parentheses is relatively ample, so the positive combination is relatively ample. This argument handles real, not just rational, perturbation coefficients. Clearing denominators handles rational H, E_i as well.

Step 2: Write $H = \sum_j a_j A_j$, with $a_j > 0$ and integral relatively ample A_j . Expand the finitely many real E_i in finitely many integral divisors; small original

coefficients give small coefficients in this larger finite list. We may therefore assume the E_i integral. Fix a rational $0 < c < a_1$ and write

$$H + \sum_i \varepsilon_i E_i = c \left(A_1 + \sum_i \frac{\varepsilon_i}{c} E_i \right) + (a_1 - c)A_1 + \sum_{j \geq 2} a_j A_j.$$

For sufficiently small ε_i , Step 1 makes the first parenthesized divisor relatively ample; the remaining terms are already positive combinations of relatively ample divisors. This proves the claim. $\square$

1.5C Nefness in the Relative Setting

We now take the relative analogue of nefness.

Definition 1.5.17 (Relative nefness). For a proper morphism $f : X \rightarrow T$, a line bundle L is f -nef if its restrictions to the geometric fibres are nef. Equivalently, in our finite-type setting one can test the inequalities

$$\int_C c_1(L) \geq 0$$

on every relative curve $C \subseteq X$. The definition extends linearly to real Cartier divisors or real line-bundle classes.

Remark 1.5.18. These definitions and the relative ampleness and nefness results also hold over algebraically closed ground fields of arbitrary characteristic. For general numerical finiteness in that setting, use the relative Néron–Severi finiteness theorem; the resolution argument given above was a proof over $\mathbb{C}$. This remark does not change the ground field of any of our already defined varieties.

Theorem 1.5.19 (Relative Kleiman theorem). *Let $f : X \rightarrow T$ be proper. If D is an f -nef real Cartier divisor, then*

$$(D^k \cdot V) \geq 0$$

for every relative irreducible subvariety V of dimension k . For every f -nef line bundle L , one likewise has

$$\int_V c_1(L)^{\dim V} \geq 0.$$

Proof. If $f(V) = t$, compute the intersection on the complete fibre X_t :

$$(D^k \cdot V) = (D_t^k \cdot V) \geq 0$$

by the absolute Kleiman theorem. Here D_t means restriction of the associated real line-bundle class, so it is defined even when the Cartier support contains the fibre. $\square$

Corollary 1.5.20. *Let $f : X \rightarrow T$ be projective, D an f -nef real divisor and H an f -ample real divisor. Then $D + \varepsilon H$ is f -ample for every $\varepsilon > 0$. Conversely, if D, H are real divisors and $D + \varepsilon H$ is f -ample for all sufficiently small $\varepsilon > 0$, then D is f -nef.*

Proof. “ $\Leftarrow$ ”: For every relative curve C ,

$$(D \cdot C) + \varepsilon(H \cdot C) > 0.$$

Let $\varepsilon \rightarrow 0$ to get $D \cdot C \geq 0$.

“ $\Rightarrow$ ”: On each projective fibre, the absolute Corollary 1.4.7 makes $D_t + \varepsilon H_t$ ample. Consequently

$$((D + \varepsilon H)^{\dim V} \cdot V) > 0$$

for every positive-dimensional relative subvariety V . For rational classes the conclusion follows from Corollary 1.5.7. For real classes we use the real version of the relative Nakai–Moishezon criterion [FM21, Theorem 1.5], which has the same intersection condition for real line bundles on proper morphisms. Thus $D + \varepsilon H$ is f -ample. $\square$

The same relative numerical criterion shows that f -ampleness of real classes depends only on relative numerical equivalence. Other relative statements below will be checked with their own hypotheses; one should not infer every absolute assertion simply by substituting relative curves.

1.5D Kleiman’s Criterion in the Relative Setting

Definition 1.5.21 (Relative ample and nef cones). For a proper $f : X \rightarrow T$, the relative ample cone $\text{Amp}(X/T) \subseteq N^1(X/T)_{\mathbb{R}}$ is the cone of f -ample real classes, and the relative nef cone $\text{Nef}(X/T)$ is the cone of f -nef classes.

The ample cone consists of finite positive combinations of relatively ample integral (or rational) classes. As in Section 1.4, we do not add 0 to it by convention; in relative dimension zero the zero class itself can be relatively ample.

Theorem 1.5.22. *Let $f : X \rightarrow T$ be projective. Then*

- (1) $\text{Nef}(X/T) = \overline{\text{Amp}(X/T)}$;
- (2) $\text{Amp}(X/T) = \text{int}(\text{Nef}(X/T))$.

Proof. The relative nef cone is closed, since its defining inequalities are closed half-spaces. Proposition 1.5.16 gives openness of relative ampleness. More explicitly, apply it to a finite set of integral classes spanning $N^1(X/T)_{\mathbb{R}}$. Thus

$$\overline{\text{Amp}(X/T)} \subseteq \text{Nef}(X/T), \quad \text{Amp}(X/T) \subseteq \text{int}(\text{Nef}(X/T)).$$

Fix an f -ample integral class H . If D is f -nef, then $D + \varepsilon H$ is f -ample for all $\varepsilon > 0$ by Corollary 1.5.20. Taking the limit gives the first reverse inclusion. If $[D] \in \text{int}(\text{Nef}(X/T))$, then $D - \varepsilon H$ is f -nef for $0 < \varepsilon \ll 1$, and

$$D = (D - \varepsilon H) + \varepsilon H$$

is f -ample by the same corollary. This is the relative version of the proof of Theorem 1.4.21. $\square$

Definition 1.5.23 (Relative cone of curves). The cone of relative curves is

$$\text{NE}(X/T) = \left\{ \sum_i a_i [C_i] : a_i \geq 0, C_i \text{ relative irreducible curves} \right\} \subseteq N_1(X/T)_{\mathbb{R}}.$$

Its closure $\overline{\text{NE}}(X/T)$ is the closed cone of relative curves.

Remark 1.5.24. The cone $\text{NE}(X/T)$ need not be closed.

Proposition 1.5.25. *The closed relative cone of curves is dual to the relative nef cone:*

$$\overline{\text{NE}}(X/T) = \{ \gamma \in N_1(X/T)_{\mathbb{R}} : (\delta \cdot \gamma) \geq 0 \text{ for every } \delta \in \text{Nef}(X/T) \}.$$

Proof. By definition and continuity, $\overline{\text{NE}}(X/T)^* = \text{Nef}(X/T)$. Apply Lemma 1.4.26 to the closed convex cone $\overline{\text{NE}}(X/T)$. $\square$

Theorem 1.5.26 (Relative Kleiman criterion). *Let $f : X \rightarrow T$ be projective and D a real divisor. Then D is f -ample if and only if*

$$\overline{\text{NE}}(X/T) \setminus \{0\} \subseteq D_{>0}.$$

For any f -ample integral divisor H and any integer k , the set

$$W_k = \{ \gamma \in \overline{\text{NE}}(X/T) : (H \cdot \gamma) \leq k \}$$

is compact. In particular it contains only finitely many numerical classes of effective relative integral one-cycles of H -degree at most k .

Proof. “ $\Rightarrow$ ”: If D is f -ample, it is non-negative on the closed cone by continuity. If $D \cdot \gamma = 0$ for some nonzero $\gamma \in \overline{\text{NE}}(X/T)$, the perfect pairing gives a real divisor E with $E \cdot \gamma < 0$. Hence $(D + tE) \cdot \gamma < 0$ for every $t > 0$. This contradicts Proposition 1.5.16, which makes $D + tE$ relatively ample for sufficiently small $t > 0$.

“ $\Leftarrow$ ”: Choose a norm and put

$$S = \{ \gamma \in \overline{\text{NE}}(X/T) : \|\gamma\| = 1 \}.$$

This is a compact subset of the unit sphere. When it is nonempty, choose bounds $D \cdot \gamma \geq a > 0$ and $H \cdot \gamma \leq b$, $b > 0$, for a fixed relatively ample integral H . Then

$$(D - \frac{a}{b}H) \cdot \gamma = D \cdot \gamma - \frac{a}{b}(H \cdot \gamma) \geq 0 \quad (\gamma \in S).$$

By scaling, $D - \frac{a}{b}H$ is f -nef, and Corollary 1.5.20 proves $D = (D - \frac{a}{b}H) + \frac{a}{b}H$ relatively ample. If S is empty, there are no relative curves: any such curve would have positive H -degree and give a point of S . Thus all fibres are zero-dimensional and f is finite. Every line bundle is then relatively ample, so the assertion holds also in this case.

For compactness, W_k is closed and is empty if $k < 0$. If it were unbounded for $k \geq 0$, choose $\gamma_i \in W_k$ with $\|\gamma_i\| \rightarrow \infty$ and pass to a subsequence such that

$$\frac{\gamma_i}{\|\gamma_i\|} \rightarrow \gamma \in \overline{\text{NE}}(X/T), \quad \|\gamma\| = 1.$$

Then

$$H \cdot \gamma = \lim_i \frac{H \cdot \gamma_i}{\|\gamma_i\|} = 0,$$

contrary to relative ampleness. Finally take an integral basis over $\mathbb{R}$ of relative numerical divisor classes. Pairing with it embeds integral relative curve classes into $\mathbb{Z}^{\rho(X/T)}$, proving their discreteness and therefore the asserted finiteness. $\square$

We finish by comparing relative classes with absolute classes when the source and target are projective.

Proposition 1.5.27. *Let $f : X \rightarrow Y$ be a morphism of projective varieties. Then:*

- (1) *Proper pushforward induces an $\mathbb{R}$ -linear map $f_* : N_1(X)_{\mathbb{R}} \rightarrow N_1(Y)_{\mathbb{R}}$.*
- (2) *$f_* \text{NE}(X) \subseteq \text{NE}(Y)$ and $f_* \overline{\text{NE}}(X) \subseteq \overline{\text{NE}}(Y)$.*
- (3) *There is a natural identification of $N_1(X/Y)_{\mathbb{R}}$ with the subspace of $N_1(X)_{\mathbb{R}}$ spanned by the contracted curves, and*

$$N_1(X/Y)_{\mathbb{R}} \subseteq \ker f_*.$$

Equality need not hold.

- (4) *Under this identification,*

$$\overline{\text{NE}}(X/Y) = \overline{\text{NE}}(X) \cap N_1(X/Y)_{\mathbb{R}}.$$

Proof. For (1), well-definedness follows from the projection formula: if $\gamma \equiv 0$ on X , then for every divisor D on Y ,

$$D \cdot f_* \gamma = f^* D \cdot \gamma = 0.$$

For (2), the pushforward of an effective cycle is effective or zero. The second inclusion follows by continuity and passage to limits.

For (3), numerical equivalence of relative cycles tests exactly the same divisors on X as absolute numerical equivalence does. This gives the injection into $N_1(X)_{\mathbb{R}}$. Every contracted curve has zero pushforward, so the inclusion in $\ker f_*$ follows. For a concrete example of strict inclusion, take an elliptic curve E and $f : E \times E \rightarrow E$ the second projection. All its fibres have the same numerical class, so $N_1(X/Y)_{\mathbb{R}} = \mathbb{R}[E \times \{0\}]$ has dimension one. The classes

$$F_1 = E \times \{0\}, \quad F_2 = \{0\} \times E, \quad \Delta$$

are linearly independent: their intersection matrix has zero diagonal and all off-diagonal entries equal to one, hence determinant two. But $f_*F_2 = f_*\Delta = [E]$, so $[\Delta] - [F_2] \in \ker f_*$ and is not a multiple of $[F_1]$. Compare [KM98, 1.46].

For (4), the inclusion from left to right is immediate. Suppose a class ξ belongs to the right-hand side but not to $K = \overline{NE}(X/Y)$. Separation gives a linear functional ℓ on $N_1(X/Y)_{\mathbb{R}}$ with $\ell \geq 0$ on K and $\ell(\xi) < 0$. Add a sufficiently small positive multiple of a relatively ample functional. The resulting functional is strictly positive on $K \setminus \{0\}$ and still negative at ξ . These conditions persist under a sufficiently small perturbation: use the compact unit slice of K for uniform strict positivity. Thus it may be chosen rational, and represented by a $\mathbb{Q}$ -Cartier divisor H on X .

The relative Kleiman criterion makes H f -ample. By Proposition 1.5.8, a sufficiently ample integral divisor M on Y makes $H + f^*M$ ample on X , after clearing denominators if needed. Since $\xi \neq 0$ lies in $\overline{NE}(X)$, absolute Kleiman gives

$$0 < (H + f^*M) \cdot \xi = H \cdot \xi + M \cdot f_*\xi = H \cdot \xi < 0,$$

a contradiction. This proves (4). □

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